

Climate Change and the Macroeconomics of Bank Capital Regulation*

FRANCESCO GIOVANARDI[†] MATTHIAS KALDORF[‡]

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Abstract

This paper studies the interactions between bank capital regulation and climate policy through the lenses of a quantitative multi-sector DSGE model with bank failure and firm default. Fossil-penalizing bank capital requirements reallocate loan supply from fossil to clean sectors and reduce aggregate emissions, but they substantially reduce liquidity provision to households and therefore have a negative welfare effect. If fossil capital requirements additionally reward investment into emission-reduction technologies, they induce a much larger emission reduction and increase welfare. However, their contribution to the net zero transition is quantitatively small compared to emission taxes. If the transition is induced by emission taxes, optimal capital requirements decline symmetrically to counteract negative aggregate loan demand effects of high emission taxes.

Keywords: Bank Capital Regulation, Macro-Banking Model, Climate Policy, Net Zero Transition, Multi-Sector Model

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*Sonny Biswas (discussant), Francesca Diluiso, Rustam Jamilov, Noemie Lisack, Fred Malherbe, Jochen Mankart, David Martinez-Miera, Christoph Meinerding, Givi Melkadze, Emanuel Moench, Valerio Nispi Landi, Martin Oehmke (discussant), Matthias Rottner and participants at the 2023 Financial Regulation Going Green (HU Berlin), Bundesbank Spring Conference, T2M (Paris), EARE (Cyprus), Developments in Business Cycle Research (Danmarks Nationalbanken), University of Cologne Energy Economics Seminar, the 2024 Conference on Frontier Risks, Financial Innovation and Prudential Regulation of Banks (Gothenburg), German Economic Association Annual Meeting (Berlin) and the NGFS Expert Network Research (Bogota) provided useful comments and suggestions. The views expressed here are our own and do not necessarily reflect those of the Deutsche Bundesbank or the Eurosystem.

[†]Prometeia & University of Cologne. Email: francesco.giovanardi@prometeia.com.

[‡]Deutsche Bundesbank, Research Centre. Wilhelm-Epstein-Str. 14, 60431 Frankfurt am Main, Germany. Email: matthias.kaldorf@bundesbank.de (Corresponding Author).

1 Introduction

Limiting the adverse effects of anthropogenic climate change to a manageable level is one of the largest challenges for economic policy in the next decades. Meaningful emission reductions require a shift away from fossil to clean energy sources. How such a sectoral reallocation - and the financial flows it generates - interacts with bank capital regulation is not well understood yet. First, it is unclear whether differential capital requirements for clean and fossil sector loans can induce a quantitatively relevant re-allocation of funds towards the clean sector, in particular when they are compared to emission taxes. Second, it remains an open question how bank capital regulation should optimally respond to a sectoral re-allocation induced by emission taxes.

This paper answers these questions through the lenses of a unified quantitative macroeconomic framework. Specifically, we extend a multi-sector DSGE model by the possibility of bank failure and default risk in the non-financial sector. Our first key finding is that even harsh fossil penalizing capital requirements fail to induce a macroeconomically relevant re-allocation towards clean energy sources. Second, capital requirements that reward abatement efforts in the fossil sector are more effective, but their impact on aggregate emissions is comparable to only a very small emission tax. Third, optimal long run capital requirements decline in the emission tax to counter the negative effect of taxes on aggregate investment, loan demand and the supply of liquid deposits to households.

Our model combines elements from the macro-banking and macro-corporate finance literature with a multi-sector production structure typically used in the assessment of climate policies at the macro level. Non-financial firms are managed by impatient firm-owners and can finance their investment with equity from households and defaultable loans from banks (Gomes et al., 2016). Banks are owned by households and financed with deposits and equity, and we allow for the possibility of idiosyncratic bank failure following Clerc et al. (2015). Households value deposits for their liquidity services and are protected from bank failure by a public bailout guarantee: depositors are repaid using taxpayer funds in the case of bank failure.¹ Together with households' valuation of liquidity services, the bailout guarantee implies that deposit financing is always cheaper than equity financing, such that the bank leverage ratio is inefficiently high. Therefore, banks are required to hold equity, which turns out to be binding in all states (Begeau, 2020). Tighter bank regulation decreases the bank failure rate but simultaneously reduces the provision of liquid deposits to households (DeAngelo and Stulz, 2015). The socially optimal bank capital requirement balances these effects.

We embed this tractable model of optimal bank capital requirements in a multi-sector production economy that features important properties of the energy sector and climate policy. Final good producers combine intermediate inputs from three different sectors that consist of a representative clean energy, fossil energy, and non-energy firm, respectively. Fossil energy firms

¹Our model purposefully focuses on the failure of individual banks and abstracts from systemic financial stability concerns. This is motivated by recent empirical (Jung et al., 2024) and quantitative macro results (Kaldorf and Rottner, 2025), who find that the effects of ambitious climate policy on systemic risk are non-negligible but manageable using standard macroprudential instruments.

generate emissions in the production process, which are taxed but can be abated at a cost. The optimal abatement effort increases in the emission tax rate (Heutel, 2012). This directly reduces emissions and is also detrimental to the return on fossil capital. In equilibrium, this return difference implies that the fossil capital share declines, reducing emissions even further.

Banks are linked to the production side through the corporate loan market. The supply of loans depends on banks' cost of capital, which in turn depend on bank capital regulation. The demand for loans is shaped by two credit frictions on the production side. On the one hand, firm managers are more impatient than households and banks, which provides an incentive to use loan financing instead of issuing equity since this allows firms to front-load dividend payouts. On the other hand, taking up loans exposes firms to default risk, as their production technology is subject to uninsurable idiosyncratic productivity shocks. Following Gomes et al. (2016), firms default on their loan obligations if their current revenues fall short of the due loan repayment. Firm default entails a restructuring cost for banks which is fully reflected in loan rates. This give rise to an endogenous capital structure choice at the firm level. The optimal take-up balances expected restructuring cost with the advantages of loan financing. Bank capital regulation affects loan supply, since it determines the amount of costly equity banks need to raise per unit of loans supplied and, thus, affects firms' funding and investment decision via the corporate loan market.

We calibrate the model to match salient empirical features of financial markets and the effects of emission taxes on the macroeconomy and across different production sectors. Using our calibration as a laboratory for different policy experiments, we first assess quantitatively to which extent bank capital regulation can act as a climate policy instrument. The insufficiently low levels of emission taxes currently in place have sparked interest in financial market instruments that can contribute to the transition to net zero. Among the most popular proposals are differentiated bank capital requirements for loans extended to clean and fossil energy firms.

Fossil penalizing capital requirements reduce the return on fossil capital by making refinancing conditions in the fossil energy sector less attractive. We use our calibration to quantitatively evaluate the effects of fossil penalizing capital requirements. The effect of setting the fossil capital requirement to its maximum of 100% is macroeconomically relevant but quite small: emissions decline by around 1% in the long run. The most important reason is that simple fossil penalizing capital requirements do not affect the share of abated emissions within the fossil sector. Conceptually, their effect on emissions resembles a (small) tax on fossil capital. At the same time, fossil penalizing capital requirements drastically reduce liquidity provision to households, such that the net welfare effect is negative.

Therefore, we allow for sustainability-linked capital requirements (SLC), which explicitly condition banks' fossil capital requirement on the abatement effort undertaken by fossil energy firms. Such a policy "engages" banks in the net zero transition, as it rewards emission reduction activities at the firm level. By contrast, the simple fossil-penalizing capital requirements resemble a policy-induced "divestment" strategy. We assume that the SLC takes on the value of 100% if fossil energy firms do not abate, and the baseline 8% capital requirement also applied to clean energy firms if fossil energy firms fully abate their emissions. Under this policy, emissions decline

by around 4.5%, but the adverse effects on liquidity provision remain large. Notably, the net welfare effect is positive and comparable to a modest emission tax. Additionally, emission taxes do not involve adverse side effects on liquidity provision to households such that bank capital regulation is generally an inferior climate policy instrument.

Why is bank capital regulation such an ineffective climate policy instrument? In the long run, the 100% fossil capital requirement implies that fossil loans are priced as if they were held directly by households. The fossil loan rate does not include banks' funding advantage stemming from the liquidity premium on deposits and the bailout guarantee. This increases the cost of debt in the fossil energy sector by approximately 100bps.² Importantly, due to the presence of financial frictions in the production sector, increasing loan rates by 100bps does not translate into a 100bps reduction in the return on fossil capital. In equilibrium, the return on fossil investment declines by merely 0.15%. By contrast, global emissions are on average taxed (or priced) at 23\$/tCO₂. This reduces the return on fossil capital by around 10%. The efficacy even of very stringent fossil capital requirements is around two orders of magnitude smaller than what is currently achieved by emission taxes and emission trading schemes. Our key results are robust to modifications in the sectoral structure of the economy, in particular the share of initially clean energy firms. We also show that neither a substantially lower cost of adopting the clean technology nor larger emission damages overturn the quantitative irrelevance of fossil penalizing capital requirements or their negative effect on welfare.

To quantify the climate impact of different regulatory policies, we subject the economy to a linear tax path that increases from zero in 2025 to 150 US-dollar per ton of carbon dioxide (\$/tCO₂) in 2050. Since this level of the emission tax implements net zero in our model, it is consistent with climate policy goals set out in the Paris agreement. The increase is unanticipated, but all uncertainty resolves immediately upon announcement. If fossil penalizing capital requirements are implemented alongside such a emission tax path, cumulated global emissions would be around 7 GtCO₂ smaller in 2050. To put this number into perspective, cumulated emissions in 2050 still amount to 99% relative to a transition path where only emission taxes increase but capital requirements stay at their initial level. It is also worth noting that global emissions amounted to 33 GtCO₂ in 2022 alone, which underscores the macroeconomically small efficacy of bank capital regulation as climate policy instrument.

We then turn to optimal bank capital regulation in response to emission taxes, which induce a re-allocation from the fossil to the clean energy sector. Crucially, the key frictions giving rise to capital requirements are between banks and depositors, i.e. the bailout guarantee and households' valuation of liquid deposits. As long as banks are sufficiently diversified across sectors, climate policy does not introduce an asymmetry into bank capital regulation. Thus, optimal capital requirements are affected by the net zero transition in so far as sectoral effects translate into aggregate effects.³ Clean, fossil, and non-energy goods are imperfect substitutes,

²For the generally small long run effects of bank capital regulation on outcomes in the production sector, we also refer to Kashyap et al. (2010).

³Studying the aggregate effects of sector-specific shocks goes back to at least Horvath (2000). To the best of our knowledge, there is no analysis of optimal bank capital regulation in multi-sector DSGE models. The optimal policy results presented in this paper, therefore, are not restricted to the effects of climate policy along the net

such that aggregate investment and aggregate loan demand contracts and banks reduce their balance sheets.⁴ This has two relevant effects on welfare and bank capital regulation. First, banks provide less liquidity to households, such that the deposit spread widens. Second, this makes deposit financing cheaper for banks, such that they increase loan supply *ceteris paribus*. This improvement in firm financing conditions (partially) mitigates the negative loan demand effect, but this comes at the cost of higher leverage ratios and default rates in the corporate sector: default risk increases symmetrically in the clean and fossil sector. The net welfare effect of deposit scarcity is clearly negative and optimal bank capital requirements decline from 8% in the baseline without emission taxes to 7.4% for emission taxes consistent with net zero.

Related Literature. Our paper is related to two fast growing strands of literature. First, we contribute to the literature on interactions between financial frictions, climate change, and climate policy. There are several theoretical results on the relevance of financial frictions for the conduct of environmental or climate policy. Heider and Inderst (2022) and Döttling and Rola-Janicka (2022) show that financial frictions affect the conduct of climate policy and the optimal Pigouvian emission tax is lower than in the absence of financial frictions. Oehmke and Opp (2022) show that green capital requirements are an ill-suited instrument to initiate a transition to net zero: preferential green capital requirements might even increase lending to brown firms if the marginal project banks finance is operated by a brown firm. On a conceptual level, the inferiority of differentiated capital requirements as climate policy instrument relative to emission taxes also relates to Davila and Walther (2022), who develop a general framework of second-best regulatory policies. The limited efficacy of green capital requirements on real effects is also in line with recent empirical work by Miguel et al. (2024).

We also contribute to the literature of financial and climate policies in quantitative E-DSGE models. A series of recent papers has however studied green-tilted central bank policies in this class of models, such as green QE (Ferrari and Nispi Landi, 2023 or Abiry et al., 2022) and green collateral policy (Giovanardi et al., 2023). Similar to our results, these papers point to a limited effectiveness of green-tilted central bank policies as climate policy instrument. Fornaro et al. (2024) propose a two-sector model with endogenous technological change and demonstrate that the investment choices in the clean sector are more sensitive to interest rates. Our model fixes the technology in each sector, such that firms in our setup would respond identically to changes in their financing conditions. By contrast, their model abstracts from financing frictions in the production sector. Taking a macroprudential angle, Carattini et al. (2023) study the effects of socially inefficient asset stranding on the macroeconomy through the lenses of a financial accelerator model. Annicchiarico et al. (2023) propose a model with climate policy and external financing constraints at the firm level, which amplifies business-cycle fluctuations, such that (green) macroprudential policy operates as stabilization instrument. In a similar setup, Diluiso et al. (2021) discuss how sector-specific capital requirements can mitigate negative shocks orig-

zero transition, but also apply to sectoral re-allocations on a more general level.

⁴Such a contraction in investment and loan demand is in line with the effects of climate policies in comparable medium scale DSGE models, see for example Coenen et al. (2024).

inating in the financial sector.

Our paper is structured as follows: Section 2 sets up the multi-sector DSGE model with two layers of default. We describe our calibration in Section 3. In Section 4, we discuss the suitability of capital requirements as climate policy instrument, discuss its relevance for the net zero transition and provide several robustness checks. The results on optimal capital requirements in response to emission taxes are presented in Section 5. Section 6 concludes.

2 Model

Time is discrete and indexed by $t = 1, 2, \dots$. The model features a representative household, banks, monopolistically competitive final good producers, a representative investment good producer, and three sectors of competitive intermediate good firms, indexed by $s \in \{c, f, n\}$. The intermediate firms produce non-energy n intermediate goods, fossil f and clean c energy goods, respectively. We assume that the fossil energy sector can choose the emission intensity of its production by paying a cost.

While both energy goods are highly (but not perfectly) substitutable, the elasticity of substitution between energy and non-energy goods is small (but not zero). The final good producer combines all intermediate goods with labor to produce the final consumption good. Investment good firms supply sector-specific investment goods. Banks raise nominal deposits from the representative household to extend nominal loans to all three intermediate good producers. The model is closed by a public sector levying emission taxes, conducting monetary policy and setting bank capital requirements.

Households. The representative household supplies n_t units of labor at the real wage w_t . The household derives utility from consumption c_t and from holding end-of-period deposits, d_{t+1} . Deposits held from time $t - 1$ to t earn the nominal interest rate r_{t-1}^D . The household's time discount factor is denoted by β . The maximization problem of the representative household is given by

$$V_t = \max_{c_t, n_t, d_{t+1}} \frac{c_t^{1-\gamma_C}}{1-\gamma_C} - \omega_N \frac{n_t^{1+\gamma_N}}{1+\gamma_N} + \omega_D \frac{d_{t+1}^{1-\gamma_D}}{1-\gamma_D} + \beta \mathbb{E}_t [V_{t+1}] \quad (1)$$

$$\text{s.t. } c_t + d_{t+1} = w_t n_t + (1 + r_{t-1}^D) \frac{d_t}{\pi_t} + \text{div}_t + T_t ,$$

where div_t collects real dividends from banks and firms. T_t is a lump sum transfer from the government. Solving the maximization problem (1) yields a standard intra-period optimality condition for labor supply $c_t^{-\gamma_C} w_t = \omega_N n_t^{\gamma_N}$ and the Euler equation for deposits

$$1 = \mathbb{E}_t \left[\Lambda_{t,t+1} \frac{1 + r_t^D}{\pi_{t+1}} \right] + \omega_D \frac{d_{t+1}^{-\gamma_D}}{c_t^{-\gamma_C}} . \quad (2)$$

Here, $\Lambda_{t,t+1} \equiv \beta \frac{c_{t+1}^{-\gamma_C}}{c_t^{-\gamma_C}}$ is the household's stochastic discount factor (sdf). We define the nominal risk-free rate r_t implied by the household sdf:

$$1 = \mathbb{E}_t \left[\Lambda_{t,t+1} \frac{1 + r_t}{\pi_{t+1}} \right]. \quad (3)$$

As deposits provide utility services to households, the deposit rate will always be below the risk-free rate and we refer to the spread $r_t^D - r_t$ as the liquidity premium. In our setup, bank deposits are nominally safe assets since depositors are protected from the risk of bank failure by public bailout guarantees, described in the next paragraph. This idea goes back at least to Gorton and Pennacchi (1990), while money in the utility specifications have been used extensively since the contribution by Poterba and Rotemberg (1986). Assuming that households value the liquidity service of bank deposits has become a commonly used feature in the macro banking literature, see also the discussion in Begenau (2020).

Banks: Technology. Banks enter period t with real liabilities from deposits issued last period $(1 + r_{t-1}^D) \frac{d_t}{\pi_t}$ and sector-specific loans l_t^s granted last period, where $s \in \{c, f, n\}$. The bank-specific realized real return on its loan portfolio is given by $\mu_t \sum_s \mathcal{R}_t^s l_t^s$ contains two components. First, it depends on sector-specific real loan payoffs that are denoted by $\mathcal{R}_t^s$ and in turn depend on (potentially) sector-specific default rates in the non-financial sector (described below). Second, banks are subject to uninsurable idiosyncratic bank risk shocks μ_t , which follow an i.i.d. log-normal distribution with a mean of one and standard deviation ς_μ . These shocks reflect unmodeled limits to diversification of loan portfolios, consistent with empirical findings by Galaasen et al. (2021).

If the idiosyncratic bank risk shock falls below a threshold level $\bar{\mu}_t$, banks are unable to service depositors. In this case, they transfer all their assets and liabilities to the public sector who covers the shortfall and pays back depositors in full while banks are protected by limited liability. This approach to modeling bank default is common in the literature, see for example Mendicino et al. (2020) and the references therein.⁵ Rationalizing bank capital requirements by a deposit insurance put or implicit bailout guarantees goes back at least to Kareken and Wallace (1978). VanHoose (2007) provides a comprehensive summary of early theories of bank capital regulation. Pennacchi (2006) demonstrates that deposit insurance is critical to bank liquidity provision but that it also creates moral hazard. Mishin (2023) provides an analysis in a DSGE

⁵Alternatively, such a bailout guarantee can be interpreted as state-dependent transfer. Bank dividends are then given by

$$div_t^b = \mu_t \sum_s \mathcal{R}_t^s l_t^s - (1 + r_{t-1}^D) \frac{d_t}{\pi_t} + S_t^{bail} + d_{t+1} - \sum_s q(\bar{m}_{t+1}^s) l_{t+1}^s.$$

If the transfer

$$S_t^{bail} = \mathbb{1}\{\mu_t < \bar{\mu}_t\} \left((1 + r_{t-1}^D) \frac{d_t}{\pi_t} - \mu_t \sum_s \mathcal{R}_t^s l_t^s \right)$$

exactly covers the shortfall, period t dividends are identical to the formulation using a transfer of bank assets to the public sector.

model that shares features with our approach. We follow Mendicino et al. (2018) in assuming that the fiscal authority incurs direct resource losses $T_t^{bail} = \zeta \cdot F(\bar{\mu}_t) \cdot \frac{d_t}{\pi_t}$ that are proportional to the amount of deposits $F(\bar{\mu}_t) \cdot \frac{d_t}{\pi_t}$ under management by the fiscal authority.

Banks: Maximization Problem The default threshold of the bank risk shock is implicitly given by the shock realization $\bar{\mu}_t$ that enables the bank to exactly repay depositors:

$$\bar{\mu}_t = \frac{(1 + r_{t-1}^D) \frac{d_t}{\pi_t}}{\sum_s \mathcal{R}_t^s l_t^s} . \quad (4)$$

We assume that banks are restructured intermediately after a failure: they can extend new loans to firms and raise equity and deposits. This facilitates aggregation into representative banks. At the end of period t , the representative bank extends loans l_{t+1}^c , l_{t+1}^f and l_{t+1}^n to clean, fossil and non-energy firms. The loan price depends on firm-specific capital structure choices $q(\bar{m}_{t+1}^s)$, described below. The period t dividend of each bank depends on its idiosyncratic shock realization μ_t and is given by

$$div_t^b = \mathbb{1}\{\mu_t > \bar{\mu}_t\} \left(\mu_t \sum_s \mathcal{R}_t^s l_t^s - (1 + r_{t-1}^D) \frac{d_t}{\pi_t} \right) + d_{t+1} - \sum_s q(\bar{m}_{t+1}^s) l_{t+1}^s .$$

Banks finance their loan portfolio with (nominal) deposits d_{t+1} or equity η_{t+1} . In the spirit of Gertler and Kiyotaki (2010), equity can be interpreted as a transfer from households. The within-period timing assumption is that loan payoffs realize and deposits from the previous period have to be repaid before the markets for new deposits and new loans open, such that banks can not raise new equity to repay liabilities from legacy deposits. Bank equity at the end of each period is implicitly defined by the following accounting identity:

$$(1 + r_t^D) \mathbb{E}_t \left[\frac{d_{t+1}}{\pi_{t+1}} \right] + \eta_{t+1} = \sum_s \mathbb{E}_t [\mathcal{R}_{t+1}^s] l_{t+1}^s , \quad (5)$$

Banks are required to finance a (potentially type-specific) fraction κ_t^s for $s = \{c, f, n\}$ of their assets with equity. When maximizing the present value of dividends, banks have to satisfy the following constraint:

$$\mathbb{E}_t \left[\frac{(1 + r_t^D) d_{t+1}}{\pi_{t+1}} \right] \leq \sum_s (1 - \kappa_t^s) \mathbb{E}_t [\mathcal{R}_{t+1}^s] l_{t+1}^s . \quad (6)$$

If there was no bank default risk and no equity requirement ($\kappa_t^s = 0$ for all s), this part of our model reduces to the setup studied in Cúrdia and Woodford (2011), in which banks are required to repay depositors in expectations. Due to the immediate restructuring assumption and the i.i.d. nature of the bank risk shock μ_t , the bank problem reduces to a two-period consideration. In Section A, we show that the capital requirement (6) is binding in all states (see also Begenau, 2020).

Banks: Loan Supply. Solving the profit maximization problem subject to the binding capital constraint (6), we obtain the loan pricing condition

$$q(\bar{m}_{t+1}^s) = \mathbb{E}_t \left[\left\{ \underbrace{(1 - \kappa_t^s) \left(\frac{\pi_{t+1}}{1 + r_t^D} - \Lambda_{t,t+1} (1 - F(\bar{\mu}_{t+1})) \right)}_{\text{Deposit Financing Wedge } \Xi_{t+1}} + \underbrace{\Lambda_{t,t+1} (1 - G(\bar{\mu}_{t+1}))}_{\text{Bank-owner sdf } \bar{\Lambda}_{t,t+1}} \right\} \mathcal{R}_{t+1}^s \right]. \quad (7)$$

Its detailed derivation is relegated to Section A. The loan pricing condition connects the expected real payoff from granting a loan to a firm in sector s to the banks' sdf, i.e. the term in curly brackets. The bank sdf in turn consists of the benefits of deposit financing Ξ_{t+1} , weighted by the deposit financed loan share $(1 - \kappa_t^s)$, and the bank-owner sdf $\bar{\Lambda}_{t,t+1}$. Bank-owners take the possibility of bank failure into account when pricing loans, which is reflected by the expected bank productivity conditional on not failing $(1 - G(\bar{\mu}_{t+1})) \equiv \int_{\bar{\mu}_{t+1}}^{\infty} \mu_{t+1} dF(\mu_{t+1})$.⁶

The deposit financing wedge shows that both the deposit insurance put and liquidity services affect the pricing of loans via bankers' sdf. The deposit financing wedge Ξ_t reflects the benefit of financing a loan through deposits $r_t^D < r_t$ due to their liquidity benefits. It also reflects the deposit insurance put: the expected repayment obligation in period $t + 1$ of raising one unit of deposits in period t is only $1 - F(\bar{\mu}_{t+1})$. All else equal, loan prices increase if capital requirements are relaxed, since this allows banks to increase the deposit financed share.

The bank-owner sdf $\bar{\Lambda}_{t,t+1}$ is closely related to the household sdf due to the perfect risk-sharing assumption, but also contains the expected bank profitability conditional on not failing, which is given by $1 - G(\bar{\mu}_{t+1})$. Since the bank risk shock has a mean of one by assumption, we have $1 - G(\bar{\mu}_{t+1}) < 1$ and the required return on bank equity is positively related to the bank failure probability. Intuitively, the bank loses control of their assets if it fails such that expected loan payoffs are discounted more heavily if bank failure is more likely. This mechanism has been referred to as "guarantee overhang" (Bahaj and Malherbe, 2020). Stringent capital requirements, therefore, also have a positive effect on loan supply by decreasing $(1 - G(\bar{\mu}_{t+1}))$, which resembles the "forced safety effect" studied by Bahaj and Malherbe (2020).

Intermediate Good Firms. We now describe the demand side for corporate loans that links bank capital regulation to climate policy in our model. There are three types of intermediate good firms which produce a homogeneous sector-specific intermediate good z_t^s for $s \in \{c, f, n\}$. The non-energy good is denoted by z_t^n , while energy goods z_t^c and z_t^f are produced by a representative clean and fossil firm, respectively. Firms are managed on behalf of impatient entrepreneurs with time preference parameter $\tilde{\beta} < \beta$. They maximize the present value of dividends, discounted by the firm-owner sdf $\tilde{\Lambda}_{t,t+1} \equiv \tilde{\beta}^{\frac{c_{t+1} - \gamma_C}{c_t}}$, which follows from the perfect risk-sharing assumption.

The production technology is linear in capital k_t^s , which can be taxed at the constant rate

⁶If banks were fully equity financed ($\kappa_t^s = 1$ for all s), the deposit financing wedge Ξ_t is irrelevant for the loan pricing condition. Furthermore, the bank failure rate would be zero, i.e. the bank-owner sdf coincides with the household sdf. Consequently, the loan pricing condition collapses to a standard consumption-based asset pricing condition $q(\bar{m}_{t+1}^s) = \mathbb{E}_t [\Lambda_{t,t+1} \mathcal{R}_{t+1}^s]$, rather than an intermediary asset pricing condition. In contrast, if there are no capital requirements ($\kappa_t^s = 1$ for all s), bank failure rate and deposit supply would be very high. Since we assume that fiscal losses during a bailout are proportional to bank deposits and the household utility function is concave in deposits, it is intuitive that there is an interior solution for κ in steady state.

τ^s . Capital has to be acquired from investment good firms that are subject to investment adjustment costs at the sector-specific price of capital ψ_t^s . We derive the supply of investment goods i_t^s as a function of its price in Section A. Intermediate good firms produce output subject to an uninsurable idiosyncratic productivity shock. As it is standard in the literature, we assume that the shock is i.i.d. log-normally distributed with standard deviation ς_M . We normalize its mean to $-\varsigma_M^2/2$, which ensures that the shock has a mean of one. To finance their investment, firms can either use equity (which reduces current dividends) or nominal long-term loans l_t^f (which reduces expected future dividends). A share $0 < \chi \leq 1$ of loans matures each period and the non-maturing share $(1 - \chi)$ is rolled over at the loan price $q(\bar{m}_{t+1}^s)$.

Firms default on the maturing share χl_t^s , if revenues from production $p_t^s m_t z_t^s$ fall below the required loan repayment χl_t^f . In this case, banks are entitled to the period t output, but have to pay a restructuring costs φl_t^s . Following Gomes et al. (2016), we assume that firms are restructured immediately which, together with the i.i.d. nature of idiosyncratic productivity shocks, permits aggregation into a representative firm in each sector. In the main text, we only present the problem of the fossil energy firm and report the first-order conditions of clean energy and non-energy firms in Section A.

Fossil Energy Firms: Technology. Fossil energy firms are subject to emission taxes τ_t^e . We follow Heutel (2012) in assuming that unabated emissions are proportional to production and allow fossil energy firms to undertake a costly abatement effort a_t . Emissions are therefore given by $e_t = (1 - a_t)z_t^f$ while the emission tax paid in period t is given by $\tau_t^e(1 - a_t)z_t^f$. Abatement costs are convex in the abatement effort and proportional to output:

$$\Theta(a_t, z_t^f) = \frac{\theta_1}{\theta_2 + 1} a_t^{\theta_2 + 1} z_t^f, \quad (8)$$

with $\theta_1, \theta_2 > 0$. The optimal abatement effort is given by

$$a_t^* = \left(\frac{\tau_t^e}{\theta_1} \right)^{\frac{1}{\theta_2}}. \quad (9)$$

The emission tax *compliance cost* per unit of fossil energy production are obtained from plugging the optimal abatement effort a_t^* back into (8) and summarizes all expenses induced by emission taxation and abatement:

$$\xi_t \equiv \tau_t^e \left(1 - \left(\frac{\tau_t^e}{\theta_1} \right)^{\frac{1}{\theta_2}} \right) + \frac{\theta_1}{\theta_2 + 1} \left(\frac{\tau_t^e}{\theta_1} \right)^{\frac{\theta_2 + 1}{\theta_2}}. \quad (10)$$

The first part of (10) summarizes the emission tax burden, while the second part are actual resource losses associated with abatement. All else equal, compliance costs increase the break-even productivity shock realization $\bar{m}_t^f$ below which the firm defaults. Banks' loan pricing condition takes this into account, such that fossil energy firms adjust their investment and leverage choices.

Fossil Energy Firms: Maximization Problem. Combining these elements, we can write dividends as

$$div_t^f = \mathbb{1}\{m_t > \bar{m}_t^f\} \cdot \left((p_t^f - \xi_t)z_t^f - \chi \frac{l_t^f}{\pi_t} \right) - \psi_t^f i_t^f + q(\bar{m}_{t+1}^f) \left(l_{t+1}^f - (1 - \chi) \frac{l_t^f}{\pi_t} \right).$$

After plugging in the law of motion for capital $i_t^f = k_{t+1}^f - (1 - \delta_K)k_t^f$, the relevant part of the maximization problem becomes

$$\begin{aligned} \max_{k_{t+1}^f, l_{t+1}^f, \bar{m}_{t+1}^f} & -\psi_t^f (1 + \tau^f) k_{t+1}^f + q(\bar{m}_{t+1}^f) \left(l_{t+1}^f - (1 - \chi) \frac{l_t^f}{\pi_t} \right) \\ & + \mathbb{E}_t \left[\tilde{\Lambda}_{t,t+1} \cdot \left\{ \int_{\bar{m}_{t+1}^f}^{\infty} (p_{t+1}^f - \xi_{t+1}) m k_{t+1}^f - \chi \frac{l_{t+1}^f}{\pi_{t+1}} dF(m) \right. \right. \\ & \quad \left. \left. + \psi_{t+1}^f (1 + \tau^f) (1 - \delta_K) k_{t+1}^f + q(\bar{m}_{t+2}^f) \left(l_{t+2}^f - (1 - \chi) \frac{l_{t+1}^f}{\pi_{t+1}} \right) \right\} \right], \end{aligned}$$

subject to the default threshold

$$\bar{m}_{t+1}^f \equiv \frac{\chi l_{t+1}^f}{\pi_{t+1} (p_{t+1}^f - \xi_{t+1}) k_{t+1}^f}, \quad (11)$$

and to banks' break-even condition on loans (7).

In the following, λ_t^f denotes the Lagrange multiplier on (11). Furthermore, the expected productivity of a defaulting firm is defined as $G(\bar{m}_t^f) \equiv \int_0^{\bar{m}_t^f} m dF(m)$ while the default probability is given by $F(\bar{m}_t^f) \equiv \int_0^{\bar{m}_t^f} dF(m)$. The risk choice $\bar{m}_{t+1}^f$ is linked to the loan price (7) through the expected real per-unit payoff:

$$\mathbb{E}_t[\mathcal{R}_{t+1}^f] = \mathbb{E}_t \left[(1 - \chi) q(\bar{m}_{t+2}^f) + \frac{\chi}{\pi_{t+1}} \left(1 - F(\bar{m}_{t+1}^f) + \frac{G(\bar{m}_{t+1}^f)}{\bar{m}_{t+1}^f} - F(\bar{m}_{t+1}^f) \varphi \right) \right]. \quad (12)$$

The first term reflects the the rollover share $(1 - \chi)$ of loans outstanding is valued at market price $q(\bar{m}_{t+1}^f)$. The second term represents the payoff from maturing share χ : it consists of the repayment probability $1 - F(\bar{m}_{t+1}^f)$, the production revenues seized in case of default $\frac{G(\bar{m}_{t+1}^f)}{\bar{m}_{t+1}^f}$, and expected restructuring cost $F(\bar{m}_{t+1}^f) \varphi$.⁷ Firms take the effect of their risk choice on loan prices through the expected loan payoff into account when making their loan and investment decisions.

Sustainability-Linked Capital Requirements. As an alternative policy that regulators can use to support the net zero transition, we consider sustainability-linked capital requirements (SLC). Here, the fossil capital requirement depends on the abatement effort a_t , such that fossil energy firms are rewarded with a lower cost of capital for undertaking abatement efforts. We

⁷The production revenues that banks seize in the default case are given by $G(\bar{m}_{t+1}^f)(p_{t+1}^f - \xi_{t+1})k_{t+1}^f$ are distributed equally among the holders of maturing loans χl_{t+1}^f . Expressing recovered revenues per unit of maturing loans, we obtain the term $\frac{G(\bar{m}_{t+1}^f)}{\bar{m}_{t+1}^f}$.

focus on a simple linear relationship:

$$\kappa_t^f = (1 - a_t) \cdot \kappa_t^{pen} + a_t \cdot \kappa_t^{reward}. \quad (13)$$

With a penalty capital requirement of $\kappa_t^{pen} = 1$, fossil loans receive a capital requirement of 100% if firms do not abate ($a_t = 0$), while they receive a capital requirement $\kappa_t^{reward} < \kappa_t^{pen}$ under full abatement. In the quantitative analysis, we set $\kappa_t^{reward} = \kappa_t^{sym}$, such that clean and fossil loans are treated equally under full abatement. Plugging (13) into (7), the loan price at which banks break even becomes:

$$q(\bar{m}_{t+1}^f) = \mathbb{E}_t \left[(1 - \kappa_t^{pen} + a_t(\kappa_t^{pen} - \kappa_t^{reward})) \Xi_{t+1} + \bar{\Lambda}_{t,t+1} \mathcal{R}_{t+1}^f \right]. \quad (14)$$

In this setting, the optimal abatement effort depends on the loan pricing condition and the abatement cost function:

$$a_t^* = \left(\frac{\tau_t^e + (\kappa_t^{pen} - \kappa_t^{reward}) \mathbb{E}_t [\Xi_{t+1} \mathcal{R}_{t+1}^f] \frac{l_{t+1}^f - (1-\chi)l_t^f}{z_t^f}}{\theta_1} \right)^{1/\theta_2}, \quad (15)$$

which collapses to (9) if $\kappa_t^{pen} = \kappa_t^{reward}$.

Intermediate Good Firm: Loan and Investment Choice. Denoting the Lagrange-multiplier on the default threshold by λ_t^f , the first-order conditions for investment and loan issuance are given by

$$q(\bar{m}_{t+1}^f) - \lambda_t^f \frac{\bar{m}_{t+1}^f}{l_{t+1}^f} = \mathbb{E}_t \left[\frac{\tilde{\Lambda}_{t,t+1}}{\pi_{t+1}} \left\{ \chi(1 - F(\bar{m}_{t+1}^f)) + (1 - \chi)q(\bar{m}_{t+2}^f) \right\} \right], \quad (16)$$

and

$$\psi_t^f - \lambda_t^f \frac{\bar{m}_{t+1}^f}{k_{t+1}^f} = \mathbb{E}_t \left[\tilde{\Lambda}_{t,t+1} \left\{ \psi_{t+1}^f (1 - \delta_K) + (p_{t+1}^f - \xi_{t+1})(1 - G(\bar{m}_{t+1}^f)) \right\} \right], \quad (17)$$

whereas the risk choice $\bar{m}_{t+1}^f$ is determined according to

$$-\lambda_t^f - q'(\bar{m}_{t+1}^f) \left(l_{t+1}^f - (1 - \chi) \frac{l_t^f}{\pi_t} \right) = \mathbb{E}_t \left[\tilde{\Lambda}_{t,t+1} \left\{ \left(l_{t+2}^f - (1 - \chi) \frac{l_{t+1}^f}{\pi_{t+1}} \right) q'(\bar{m}_{t+2}^f) \frac{\partial \bar{m}_{t+2}^f}{\partial \bar{m}_{t+1}^f} \right\} \right]. \quad (18)$$

The first-order condition for loans (16) equates the marginal benefit of taking up a loan with its costs. Since the loan maturity exceeds one period, it also contains a dilution term that enters through the multiplier on the default threshold λ_t^f , which is in turn pinned down by equation (18). The cost of marginally increasing loans consist of the redemption share χ , weighted by the repayment probability, and the roll-over part $(1 - \chi)$, valued by next period's loan price $q(\bar{m}_{t+2}^f)$. Equation (17) requires that the after-tax cost of investment $(1 + \tau^f)\psi_t^f$ equals its expected discounted payoff, which consists of the value of undepreciated capital next period and expected revenues per unit of capital, conditional on repayment. Per-unit revenues depend on

the fossil energy price net of emission taxes and abatement costs, summarized by the compliance cost ξ_{t+1} . Since investment also affects the default probability in future periods, the multiplier on the risk choice also enters the first-order condition for investment. Section A contains the analytical expression of the loan price's derivative with respect to the risk choice as well as an additional condition that pins down the multiplier λ_t^f on the default risk threshold.

Final Good Firms. Monopolistically competitive firms aggregate both energy inputs and the non-energy intermediate good together with labor into the final good y_t according to a nested CES-structure (see also Fried et al., 2022):

$$y_t = A_t(1 - \mathcal{D}_t)\tilde{z}_t^\alpha n_t^{1-\alpha}, \quad (19)$$

where exogenous total factor productivity A_t follows $\log(A_t) = \rho_A \log(A_{t-1}) + \sigma_A \epsilon_t^A$ and $\mathcal{D}_t$ are damages associated with climate change, described below. The parameter α governs the labor share in the aggregate production function and the intermediate goods bundle is defined as

$$\tilde{z}_t = \left(\tilde{\nu}(z_t^e)^{\frac{\tilde{\epsilon}-1}{\tilde{\epsilon}}} + (1 - \tilde{\nu})(z_t^n)^{\frac{\tilde{\epsilon}-1}{\tilde{\epsilon}}} \right)^{\frac{\tilde{\epsilon}}{\tilde{\epsilon}-1}}, \quad (20)$$

where $\tilde{\nu}$ is the weight on energy in the intermediate goods bundle and $\tilde{\epsilon}$ is the elasticity between energy and non-energy goods. The energy bundle is, in turn, given by

$$z_t^e \equiv \left(\nu(z_t^c)^{\frac{\epsilon-1}{\epsilon}} + (1 - \nu)(z_t^f)^{\frac{\epsilon-1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon-1}}, \quad (21)$$

where ν is the clean energy weight in the energy bundle. Solving the profit maximization problem yields standard demand functions for labor and all intermediate goods, shown in detail in Section A. Each firm i sells its final good variety to households at a markup over its marginal costs, subject to price adjustment costs a la Rotemberg (1982). This gives rise to a New Keynesian Phillips curve linking monetary policy to the production sector. We relegate the analytical steps to Section A.

Public Sector and Resource Constraint. The model is closed by assuming that emission tax revenues and losses from bank bailouts are rebated in lump-sum fashion to households ($T_t = \tau_t^e e_t - T_t^{bail}$) and that the central bank sets the nominal interest rate according to a Taylor-type rule:

$$1 + r_t = (1 + r^{SS})\pi_t^{\varphi_\pi} + \sigma_R \epsilon_t^R. \quad (22)$$

The resource constraint is given by

$$y_t = c_t + \sum_s i_t^s \left(1 + \frac{\Psi_I}{2} \left(\frac{i_t^s}{i_{t-1}^s} - 1 \right)^2 \right) + \frac{\Psi_P}{2} (\pi_t - 1)^2 + \frac{\theta_1}{\theta_2 + 1} \left(\frac{\tau_t^e}{\theta_1} \right)^{\frac{\theta_2 + 1}{\theta_2}} z_t^f + \varphi F(\bar{m}_t) + \zeta F(\bar{\mu}_t) d_t, \quad (23)$$

where the abatement costs are evaluated at the optimal abatement effort a_t^* , given the emission tax τ_t^e . Note that the benefits of higher deposit supply do not enter the resource constraint, but are part of the welfare objective via the household utility function. Climate change damages are linked to emissions:

$$\mathcal{D}(E_t) = 1 - \exp\{\psi_E E_t\}, \quad (24)$$

where the parameter ψ_E governs the damage-to-GDP ratio.

3 Calibration

The model is calibrated to financial market data from the US and each period corresponds to one year. All data moments are based on the sample from 1984 to 2013. While historical bank failure rates are publicly available only after 1984, the sample truncation point in 2013 implies that our data sample does not cover the period after the Paris agreement, which was signed in 2015. This justifies a symmetric calibration of all parameters related to intermediate good producers. Consequently, we set the emission tax to zero when computing model-implied moments during our calibration and also assume that the capital requirement is identical across sectors ($\kappa_t^c = \kappa_t^f = \kappa_t^n = \kappa^{sym}$). We describe parameters associated with each group of agents separately.

Households. We fix household's consumption CRRA parameter to $\gamma_C = 2$, the labor disutility curvature $\gamma_N = 1$, and the weight on labor disutility to $\omega_N = 8$ to imply a steady state labor supply $n^{SS} = 0.3$. The household discount factor is set to $\beta = 0.99$, implying an annualized real interest rate of 1%. The elasticity and weighting parameters in household's valuation of liquidity services are calibrated to match the excess volatility of deposit holdings over GDP, and the liquidity premium, defined as the average deposit spread below the risk-free rate $r_t^D - r_t$. The ratio of liquid assets to GDP can be computed from the data as in Lucas and Nicolini (2015) and the excess volatility of this ratio over the volatility of GDP is given by 0.98. We target a liquidity premium of -40bps (van Binsbergen et al., 2022; Krishnamurthy and Li, 2022). The resulting weighting parameter is given by $\omega_D = 0.007$, while the curvature is set to $\gamma_D = 1.5$.

Banks. Closely related to households valuation of liquidity services, the next set of parameters concerns banks. Historical bank failure rates are computed based on data from the Federal Deposit Insurance Corporation (FDIC). Following Elenov et al. (2021), we compute the bank failure rate as the share of assisted mergers and banks in resolution of all banks. This yields an average bank failure rate of 0.7% per year, which obtains in our model by setting the standard deviation of banks' risk shock to $\varsigma_\mu = 0.0325$.⁸ We set the deadweight loss parameter of the deposit insurance agency to $\zeta = 0.02$, which renders a symmetric long run capital requirement of 8% optimal in the Ramsey policy problem.

⁸The same data moment is also used by Mendicino et al. (2020) in a calibration to euro area data.

Parameter	Value	Source
<i>Households</i>		
Household discount factor β	0.99	Standard
Consumption CRRA γ_C	2	Standard
Liquidity curvature γ_D	1.5	Target: Excess vol. deposit/GDP
Liquidity weight ω_D	0.008	Target: Liquidity premium
Labor supply curvature γ_N	1	Frisch elasticity of one
Labor supply weight ω_N	8	$n^{SS} = 0.33$
<i>Technology</i>		
Inv. adj. parameter Ψ_I	2	Target: Excess vol. investment
Capital depreciation rate δ_K	0.08	Standard
Cobb-Douglas coefficient α	1/3	Standard
Energy weight $\tilde{\nu}$	0.0015	Energy share 10%
Energy/non-energy CES $\tilde{\epsilon}$	0.2	Bartocci et al (2024)
Clean weight ν	0.3865	Fossil energy share 80%
Fossil/clean energy CES ϵ	3	Fried et al (2022)
Abatement cost parameter θ_1	0.05	Full Abatement Tax 150\$/tCO2
Abatement cost parameter θ_2	1.6	Heutel (2012)
Emission damage parameter Ψ_E		Damage/GDP ratio
<i>Financial Markets</i>		
Firm-owner discount factor $\tilde{\beta}$	0.988	Target: Firm leverage
St. dev. firm risk ς_m	0.2	Target: Firm default rate
Loan maturity parameter χ	0.2	5-year average maturity
St. dev. bank risk ς_μ	0.0275	Target: Bank failure rate
Restructuring costs φ	0.15	Target: Recovery rate
Bailout loss ζ	0.027	Optimality of κ^{sym}
Capital requirement κ^{sym}	0.08	Basel III
<i>Nominal Rigidities</i>		
Price adjustment cost Ψ_P	4.5	Standard
Final good CES ϕ	6	20% Markup
Taylor rule coefficient φ_π	2	Target: Excess vol. inflation
<i>Shocks</i>		
MP shock st. dev. σ_R	0.02	Target: Excess vol. nominal rate
Persistence TFP ρ_A	0.8	Standard
TFP shock st. dev. σ_A	0.004	Standard

Table 1: Baseline Calibration

Intermediate Good Firms. The parameters governing financial frictions of intermediate good firms are set to match moments typically used in the macro-finance literature. We set the annual capital depreciation rate to a standard value of $\delta_K = 0.08$ and the average loan maturity to five years, implying $\chi = 0.2$. We jointly calibrate the standard deviation of idiosyncratic productivity shocks $\varsigma_M = 0.2$, the firm owner discount factor $\tilde{\beta} = 0.988$ and the restructuring cost parameter $\varphi = 0.12$ to match the corporate default rate, the leverage ratio, and the recovery

rate on loans. In the model, the expected recovery rate per unit of loans is given by the realized payoff from holding a distressed loan relative to the promised payoff. It is related to the expected productivity of a defaulter, i.e. the expected productivity conditional on defaulting $\frac{G(\bar{m}_t^s)}{F(\bar{m}_t^s)}$ and the restructuring cost φ , i.e.

$$recov(\bar{m}_t^s) = \frac{G(\bar{m}_t^s)}{F(\bar{m}_t^s)\bar{m}_{t+1}^s} - \varphi. \quad (25)$$

Our parameterization implies recovery rate of 82%, which corresponds to the average debt recovery rate for the US, based on the dataset provided by Djankov et al. (2008). We target a corporate default rate of 1.7% and a leverage ratio of 30% at book values, which is given by $lev_t^s = l_t^s/k_t^s$ in our model. The data moments correspond to the average default rate of rated firms and to the ratio of corporate sector net debt over net assets in the US from 1984-2013, respectively.

Final Producers. The Cobb-Douglas coefficient in the production function of final good producers is fixed at $\alpha = 1/3$. The elasticity of substitution between final goods varieties $\phi = 6$ implies a markup of 20%. We set the Rotemberg price adjustment parameter to $\Psi_P = 4.5$, which implies an average duration of five quarters. The central bank reaction to inflation $\phi_\pi = 2$ and the standard deviation of monetary policy shocks $\sigma_R = 0.02$ matches the excess volatility of inflation and short-term interest rates over GDP. The investment adjustment cost parameter is set to $\Psi_I = 2$, targeting the excess volatility of investment over GDP. Lastly, persistence and standard deviation of the aggregate TFP shock are set to $\rho_A = 0.8$ and $\sigma_A = 0.004$, which are standard values in the business cycle literature. Since we only target excess volatilities of endogenous objects over the volatility of GDP, the parameters of the TFP shock are not central to our calibration exercise. We have nine free parameters related to macroeconomic fluctuations, monetary policy, and financial frictions, which we set to jointly match a set of five time series means and four excess volatilities. Naturally, each parameter also affects all other moment-implied moments, but our model allows us to quite clearly associate one moment to each parameter that is particularly strongly affected.

Sectoral Shares. The parameters governing the sectoral structure of the production side can be set match sectoral shares in the initial steady state. This follows from the assumption of symmetric financial and investment adjustment frictions and the absence of sector-specific shocks, which together implies that the sectoral shares are constant over time. The elasticity $\tilde{\epsilon} = 0.2$ between energy and non-energy goods in equation (20) follows Bartocci et al. (2024) who calibrate a medium scale DSGE model to sectoral data from the EU. The weighting parameter $\tilde{\nu} = 0.0015$ then implies an energy share of 10% in the final good production, which is also used in Bartocci et al. (2024). Within the energy sector, we set $\nu = 0.3865$ implying that 20% of firms in the energy sector use the clean technology initially. As in Fried et al. (2022), we fix the elasticity between clean and fossil energy at $\epsilon = 3$ in equation (21).

Moment	Model	Data
<i>Targeted Means</i>		
Liquidity premium $ave(s^D)$ in bps	-38	-40
Corporate leverage $ave(lev)$ in %	29	26
Corporate default $ave(F(\bar{m}))$ in %	1.6	1.7
Recovery rate $ave(recov)$ in %	81	82
Bank failure rate $ave(F(\bar{\mu}))$ (%)	0.6	0.6
<i>Targeted Excess Volatilities</i>		
Relative vol. investment $\sigma(i)/\sigma(y)$	1.72	1.50
Relative vol. deposit/GDP $\sigma(d/y)/\sigma(y)$	0.98	1.02
Relative vol. interest rate $\sigma(r)/\sigma(y)$	0.70	0.98
Relative vol. inflation $\sigma(\pi)/\sigma(y)$	0.87	0.68
<i>Untargeted Excess Volatilities</i>		
Relative vol. consumption $\sigma(c)/\sigma(y)$	0.83	0.88
Relative vol. leverage $\sigma(lev)/\sigma(y)$	0.50	0.93
Relative vol. corporate default $\sigma(F(\bar{m}))/\sigma(y)$	0.50	0.86
Relative vol. bank failure $\sigma(F(\bar{\mu}))/\sigma(y)$	0.26	0.33
<i>Untargeted Correlations with GDP</i>		
Emissions-GDP $cor(y, e)$	0.64	0.78
Deposit/GDP-GDP $cor(y, d/y)$	-0.64	-0.22
Leverage-GDP $cor(y, lev)$	-0.78	-0.58
Firm default-GDP $cor(y, F(\bar{m}))$	-0.70	-0.32
Bank failure-GDP $cor(y, F(\bar{\mu}))$	-0.66	-0.32
Firm default-Bank failure $cor(F(\bar{m}), F(\bar{\mu}))$	0.58	0.56
<i>Untargeted Correlations with Inflation</i>		
Firm default-Inflation $cor(\pi, F(\bar{m}))$	-0.60	-0.33
Bank failure-Inflation $cor(\pi, F(\bar{\mu}))$	-0.25	-0.11

Table 2: Model Fit: All model-implied moments are computed based on a second-order approximation around the deterministic steady state. All moments are expressed in relative deviations from their steady state value. All data moments are based on real data and de-trended using an HP-filter with smoothing parameter of 6.25.

Climate Block. The weighting and curvature parameters of abatement costs are set to $\theta_1 = 0.05$ and $\theta_2 = 1.6$. While the curvature parameter is a standard value in the literature (Heutel, 2012), we inform θ_1 using the decline in global emission intensities since 1990, which can directly be mapped into the abatement share a_t in our model. Specifically, the emission intensity declined by 33% over the past 30 years. At the same time, global average emission taxes average at around 18\$/tCO2 over the past ten years. We use this information to back out $\theta_1 = 0.05$: in this case, an emission tax of $\tau^e = 0.0085$, which corresponds to a 18\$/tCO2 emission price, induces the empirically observed decline in emission intensities.⁹ Under this parameterization, around two percent of GDP is spent on abatement under the net zero emission tax, consistent with

⁹The model-implied emission tax is converted into \$/tCO2 by converting model units of output ($y^{model} = 0.403$ in the steady state without emission taxes) to global GDP ($y^{world} = 105.00$ billion USD in 2022) and convert model emissions ($e^{model} = 0.306$ in the baseline calibration) into global emissions ($e^{world} = 37.5$ GtCO2 in 2022).

commonly used Integrated Assessment Models (IAM). This parameterization of the abatement cost function implies full abatement for any emission tax exceeding 105\$/tCO₂. Our baseline parameterization is summarized in Table 1.

The upper two panels of Table 2 presents the model fit to targeted first and second moments. The third panel demonstrates the model’s ability to reconcile additional untargeted second moments typically used in the macro-finance and macro-banking literature. Consumption, leverage, corporate default and bank failure are all less volatile than GDP, consistent with the data. The third and fourth panel consider correlations of different macro-finance aggregates with GDP and inflation. The pro-cyclicality of emissions is captured well by the model. Specifically, the model implies a counter-cyclical default probability in the corporate sector, which plays a crucial role for dynamic bank capital regulation. Similarly, the bank failure rate inherits the counter-cyclicality of corporate default. Consequently, bank failure and corporate default are positively correlated. Lastly, default risk is generally negatively associated with inflation due to the nominal denomination of deposits and loans.

4 Bank Regulation as Climate Policy Instrument

In this section, we use our calibrated model to study the implications of differentiated capital requirements as climate policy instrument. In Section 4.1, we compare the long run implications of different climate policy instruments before providing intuition for such small effects in Section 4.2. In Section 4.3, we show how differentiated bank capital requirements affect the transition to net zero. Section 4.4 shows that our results are robust to variations in the climate block in the sectoral structure of the production side.

4.1 Long Run Effects

Figure 1 summarizes how tilted capital regulation (on the x-axis) affects key welfare-relevant aggregates in the long run. The dashed red line refers to simple fossil penalizing capital requirement, while the dotted green line corresponds to the penalty capital requirement κ_t^{pen} in the SLC (13). The top left panel shows that only the SLC induces abatement incentives. Consequently, fossil capital requirements only have a modest impact on emissions at the macroeconomic level, which are around 1% smaller than in the baseline without policies, see the top right panel. By contrast, emissions decline by almost 5% for the SLC. Further details on the sectoral effects on these policies are outlined in Figure C.1.

In addition to emissions, bank capital requirements also affect other welfare-relevant macroeconomic aggregates. Requiring banks to hold more equity for fossil loans makes banks safer but also reduces the supply of deposits to households. This makes deposits more valuable and the deposit spread over the risk-free rate widens, see the bottom row of Figure 1, and reduces welfare. At the same time, a lower bank failure rate has a positive welfare effect through the reduction

From the expression $p^e \cdot \frac{e^{\text{world}}}{y^{\text{world}}} = \tau^e \cdot \frac{e^{\text{model}}}{y^{\text{model}}}$, the model-implied emission price is given by $p^e = \frac{y^{\text{world}}/y^{\text{model}}}{e^{\text{world}}/e^{\text{model}}} \tau^e$ in \$/tCO₂.

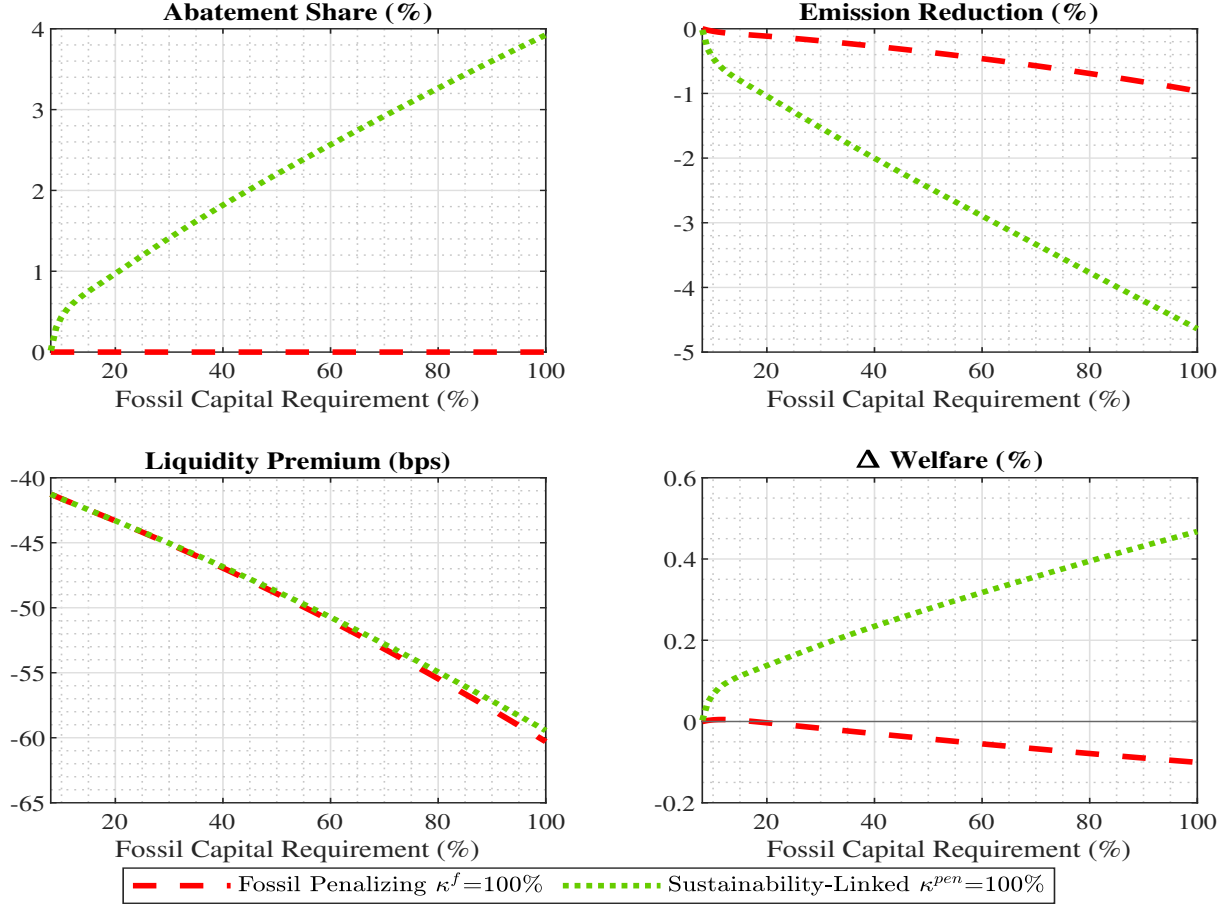


Figure 1: Long Run Effects of Fossil Capital Requirements. The upper left panel displays the share of abated emissions a_t for different κ_t^f (dotted red line) and different penalty requirements κ_t^{pen} , while fixing $\kappa_t^{reward} = \kappa_t^c = 0.08$. The liquidity premium $r_t^D - r_t$ is annualized. Welfare is defined in Equation (1), computed relative to the baseline without climate policies and expressed in consumption equivalents.

of deadweight losses from restructuring banks. The net welfare effect, which takes into account consumption, liquidity supply and the damages avoided by reducing emissions, is shown in the bottom right panel.

The welfare effect of fossil penalizing capital requirements is positive but quantitatively negligible for κ_t^f between 8% and 15%. Such a policy improves the economy's environmental performance by the fossil energy sector, similar to a direct tax on fossil capital. This resembles a policy-induced "divestment" strategies, as fossil energy firms have to finance a larger share of their investment using costly equity. For larger fossil capital requirements, the adverse effects on liquidity supply dominate the net welfare effect, which is negative for κ_t^f exceeding 15%. By contrast, sustainability-linked capital requirements increase total welfare, since the emission reduction associated with a higher abatement share has a strong positive effect. The effect of such a policy-induced "engagement" strategy clearly exceeds the negative liquidity supply effects. Notably, liquidity supply also contracts by less, since fossil loans do not receive a 100% capital requirement, but can be partially deposit financed, even though this share is very small, i.e. below the abatement share $a_t < 5\%$.

To put the effects of tilted bank capital regulation into perspective, is helpful to benchmark

tilted bank capital requirements against different fiscal instruments. The second column of Table 3 summarizes the implications of increasing the capital requirement on fossil loans to $\kappa^f = 1$, which corresponds to the most restrictive policy available to bank regulation. Compared to the baseline model, the liquidity premium widens by almost 20bps and the bank failure probability is essentially zero. The climate policy effect is very limited and comparable to the effects of a simple 2% tax on fossil capital, which we display in the fifth column of Table 3. Such a tax yields a comparable emission reduction but also does not affect the emission intensity. Therefore, the fossil energy share declines slightly, but the emission intensity of fossil energy firms remains unchanged. While the tax on fossil capital does not have a long run effect on bank failure rates and the supply of deposits it still reduces welfare by distorting investment incentives, which is a standard feature of capital taxes in such models.

Moment	Baseline	$\kappa^f = 100\%$	SLC	$\tau^e = 3\$/\text{tCO}_2$	$\tau^f = 2\%$
Fossil Energy Share	80.00%	79.17%	76.11%	72.36%	79.03%
Δ Emissions	-	-0.96%	-4.64%	-9.64%	-2.12%
Bank Failure Prob	0.54%	<0.1%	<0.1%	0.54%	0.54%
Liquidity Premium	-42bps	-60bps	-59bps	-42bps	-41bps
Welfare Gain	-	-0.10%	+0.48%	+1.19%	-0.17%

Table 3: Long Run Effects of Selected Policies: We compare the baseline model without emission taxes and symmetric capital requirements (column 1) to different climate policies. Column 2 refers to a simple fossil-penalizing capital requirement ($\kappa^f = 1$), column 3 to a sustainability-linked capital requirement with $\kappa^{pen} = 1$ and $\kappa^{reward} = 0.08$, column 4 to a $3\$/\text{tCO}_2$ emission tax, and column 5 to a 2% tax on fossil capital. The emission change and welfare are computed relative to the baseline. The fossil energy share is defined relative to total energy production $(1 - a_t^*)z_t^f/z_t^e$. The bank failure rate and the liquidity premium $r_t^D - r_t$ are annualized. Welfare is defined in Equation (1), computed relative to the baseline without climate policies and expressed in consumption equivalents.

The third column of Table 3 displays the effects of sustainability-linked capital requirements. Without emission taxes in place, the optimal abatement effort turns out to be close to 4%, such that banks have to finance around 96% of fossil loans using costly equity. This still eliminates bank failure and implies a drastic reduction in liquidity provision, which is reflected in the liquidity premium of -59bps. The sectoral re-allocation is slightly smaller than in the case of a simple fossil penalizing factor, but emissions are 4.5% smaller. This suggests that sustainability-linked capital requirements are a more powerful climate policy instrument than simple penalizing factors. Nevertheless, their effect on environmental outcome is comparable to a modest emission tax τ^e . In the fourth column, we show that a emission tax of $3\$/\text{tCO}_2$ delivers a much larger emission reduction of almost 10% without inflicting visible changes in deposit supply of bank failure. Consequently, the welfare increase is considerably larger.

4.2 Discussion

Why is the effect of fossil penalizing capital requirements on the long run fossil energy share so small in comparison to emission taxes? The transmission of emission taxes operates through the first-order condition for fossil investment in a straightforward way. By contrast, the transmission mechanism of bank regulation to the real economy operates through the loan pricing condition,

which in turn enters the investment decision across sectors. Higher fossil capital requirements increase the share of fossil loans financed by costly equity. Banks (partially) pass on their higher cost of capital to firms, i.e. credit supply contracts, which reduces the amount of funds fossil energy firms obtain by taking up loans. To gain intuition how such a credit supply contraction translates into a small equilibrium capital re-allocation across sectors, it is helpful to consider a simplified version of the first-order condition for fossil capital.

Throughout this section, we focus on the long run by setting $A_t = 1$ for every t , and we drop the expectations operator with respect to aggregate variables. To obtain analytically tractable first-order conditions, we consider the case of one period debt ($\chi = 1$) and assume that there are no debt restructuring cost ($\varphi = 0$) but that all output is lost in the case of default. Furthermore, there is full capital depreciation ($\delta_K = 1$) and no investment adjustment cost ($\Psi_I = 0$), such that the price of capital satisfies $\psi_t^f = 1$. Optimal investment is then given by

$$1 + q'(\bar{m}_{t+1}^f)(\bar{m}_{t+1}^f)^2(p_{t+1}^f - \xi_{t+1}) = \tilde{\beta} \left((p_{t+1}^f - \xi_{t+1})(1 - G(\bar{m}_{t+1}^f)) \right). \quad (26)$$

The simplified first-order condition (26) requires that the cost of purchasing one unit of capital equals its expected payoff. Fossil investment depends on the fossil energy price p_{t+1}^f , on carbon taxes through the compliance cost term ξ_{t+1} , and on bank capital requirements through several terms related to loan market frictions. As we are interested in the equilibrium fossil energy share and how it depends on emission taxes and fossil-penalizing capital requirements, we need to link this condition to the demand for fossil energy, which is an additional equilibrium condition connecting the fossil energy price p_{t+1}^f to fossil energy production z_t^f . We furthermore abstract from labor, non-energy goods and monopolistic competition in the aggregate production function. In Appendix A.1, we show that the equilibrium fossil energy share can be approximately expressed as

$$1 \approx \tilde{\beta} \left\{ \left((1 - \kappa_t^f) \Xi_{t+1} + \beta \right) f(\bar{m}_{t+1}^f)(\bar{m}_{t+1}^f)^2 + 1 \right\} \left((1 - \nu) \left(\frac{y_t}{z_t^f} \right)^{1/\epsilon} - \xi_{t+1} \right). \quad (27)$$

The effect of fossil capital requirements and emission taxes on the equilibrium fossil energy share is summarized in the single, easily interpretable equation (27). Irrespective of financial frictions in the production sector, a higher emission tax increases the climate policy compliance cost ξ_{t+1} on the left side of equation (27), such that the fossil energy share declines.¹⁰

Now consider fossil capital requirements, i.e. the term in curly brackets. Setting $\kappa^f = 1$ eliminates the deposit financing wedge in the loan price and debt financing for fossil energy firms is as expensive as possible. This would correspond to the case of households holding all

¹⁰Note that the term in curly brackets drives a wedge in the relationship between emission taxes and the fossil energy share. Quantitatively, it turns out that this wedge is quantitatively negligible for the transmission of emission taxes. In the special case where investment is fully equity financed, i.e. $\bar{m}_{t+1}^f = 0$. Here, the the equilibrium fossil capital share can be written as

$$\frac{1}{\tilde{\beta}} + \xi_{t+1} = (1 - \nu) \left(\frac{y_t}{z_t^f} \right)^{1/\epsilon}. \quad (28)$$

loans directly. Quantitatively, eliminating the deposit financing wedge approximately equals the sum of the bank failure rate and the liquidity premium, which adds up to almost 100 basis points. However, corporate default risk implies that this 100 basis point increase in firms cost of debt does not translate into a 100 basis point increase in the required return on fossil capital, which would be quite substantial. With corporate finance frictions, the deposit financing wedge is multiplied by the expression $f(\bar{m}_{t+1}^f)(\bar{m}_{t+1}^f)^2$. This expression measures the relevance of loan conditions for firm investment. Note that $\bar{m}_{t+1}^f$ is related to the leverage ratio and the share of debt that has to be rolled over each period, see equation (11). If firms would be completely equity financed, $\bar{m}_{t+1}^f$ is zero and fossil capital requirements would have no effect at all.

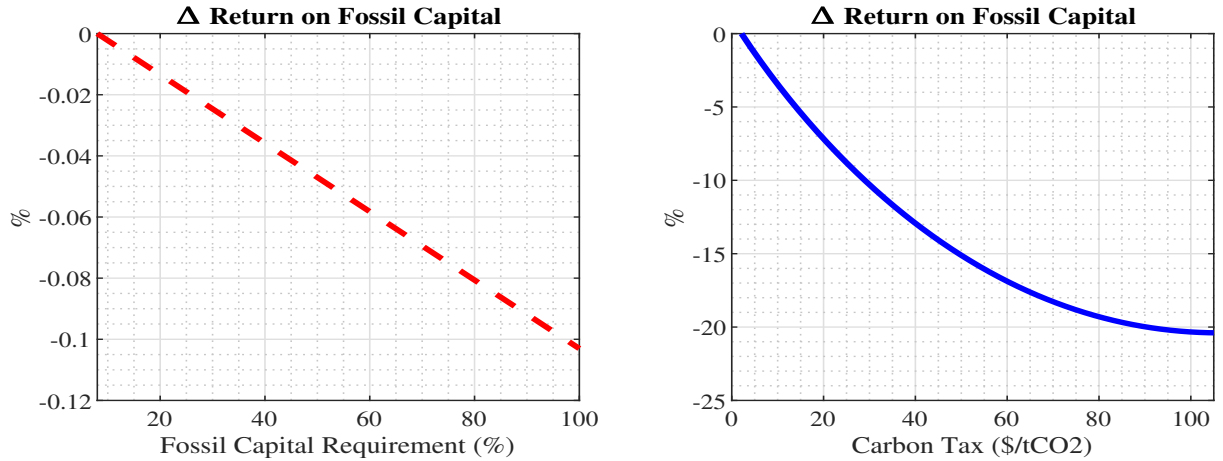


Figure 2: Fossil Requirements versus Emission Taxes. The left panel displays the return on capital wedge induced by fossil capital requirements in (27) for different κ^f (dotted red line), relative to the energy price p_t^e , fixing the emission tax at $\tau^e = 0$. The right panel display the return on capital wedge induces by emission taxes in (27) for different τ^e , fixing $\kappa^f = 0.08$. All variables are long run time series means.

The extent to which a change in the cost of debt affects the fossil investment share is, therefore, a quantitative one. In our baseline calibration, the expression related to corporate default risk amounts to $f(\bar{m}_{t+1}^f)(\bar{m}_{t+1}^f)^2 \approx 0.135$. Increasing the fossil capital requirement from 8% to 100% elevates the long run cost of capital in the fossil sector by around $(1 - 0.08) \cdot 0.135 \approx 10$ basis points. The dashed red line in the left panel of Figure 2 shows how larger fossil capital requirements affect the return on capital. Contrasting this with the effects of emission taxes in the right panel of Figure 2, it stands out that emission taxes are more than two orders of magnitude more powerful.

The transmission mechanism of sustainability-linked capital requirements additionally depends on the abatement effort (9). Here, it stands out that emission taxes and the SLC enter the first-order condition for abatement in an additive fashion. The abatement effort induced by sustainability-linked capital requirements depends positively on the policy parameters $\kappa^{pen} - \kappa^{reward}$ and also on the expression $\frac{l_{t+1}^f - (1-\chi)l_t^f}{z_t^f}$, which is again linked to financial frictions in the production sector. In steady state, this expression reduces to $\frac{\chi l^f}{k^f}$, which is related to the share of loans χ that has to be rolled over each period and the relevance of debt financing conditions, i.e. the leverage ratio $lev^f \equiv \frac{l^f}{k^f}$.

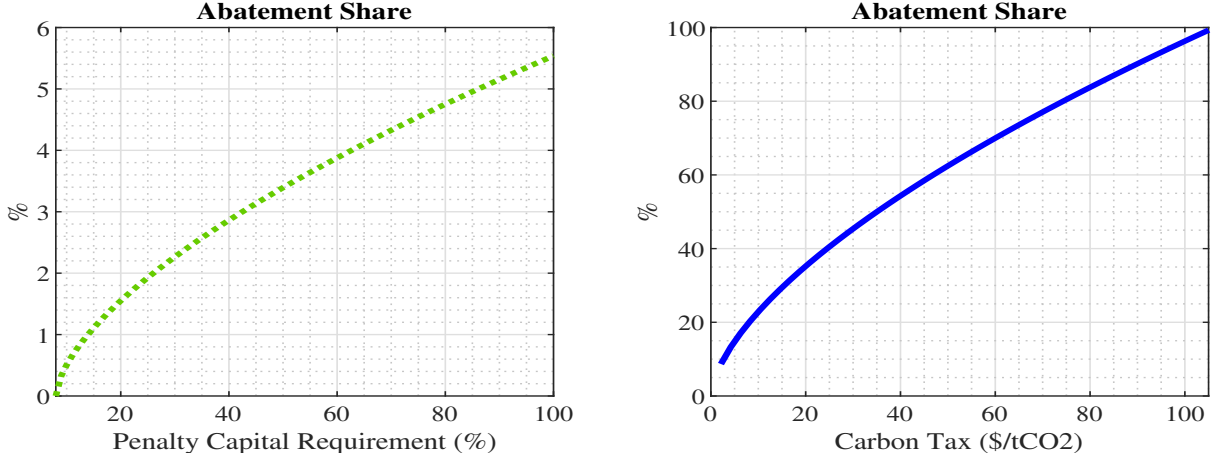


Figure 3: SLC versus Emission Taxes. The left panel displays the share of abated emissions a^* for different κ^f (dotted red line) and different penalty requirements κ^{pen} , while fixing $\kappa^{reward} = \kappa^c = 0.08$ and $\tau^e = 0$. The right panel displays the share of abated emissions for different emission taxes, without an SLC. All variables are long run time series means.

How much additional abatement effort is induced by the SLC in comparison to emission taxes is again a quantitative question. The left panel of Figure 3 shows the results from varying κ^{pen} while keeping κ^{reward} at 8% and the emission tax at zero. Consistent with the results in Table 3, the abatement share is about 5% for the most restrictive policy. The same abatement share could also be implemented with emission taxes close to 5\$/tCO₂ and misses net zero by a factor of around 20, as the right panel of Figure 3 illustrates.

4.3 Transition Dynamics

Lastly, we compute the environmental impact of differential capital requirements when they support a net zero transition induced by emission taxes. We solve and simulate the model under the assumption that emission taxes are increased linearly from zero to 105\$/tCO₂. We interpret 2025 as the start year and assume that the terminal tax is reached in 25 years. This path is consistent with climate policy goals set out in the Paris agreement: a 105\$/tCO₂ tax implements net zero in our calibration. The shift towards a more stringent climate policy is unanticipated but all uncertainty about its path is resolved immediately.

The black line in Figure 4 shows cumulated emissions from 2025 until 2050, which follow a concave shape due to the negative effect of higher emission taxes on future emissions. If bank regulation additionally imposes a 100% capital requirement on fossil loans (dashed red line) or sustainability-linked capital requirements (dotted green line), emissions are slightly smaller. Compared to the case of emission taxes only, cumulated emissions are around 7 GtCO₂ smaller in 2050 if a 100% fossil capital requirement is in place. Sustainability-linked capital requirements provide a reduction of merely 5.5 GtCO₂.

The reason behind this at first glance puzzling result is that an SLC quickly loses its relevance. The effective capital requirement (13) declines once fossil energy firms choose a large abatement share due to the high emission tax. This then implies a smaller loan rate and enables them to accumulate capital, similar to the case of emission taxes only. In contrast, a fossil penalizing

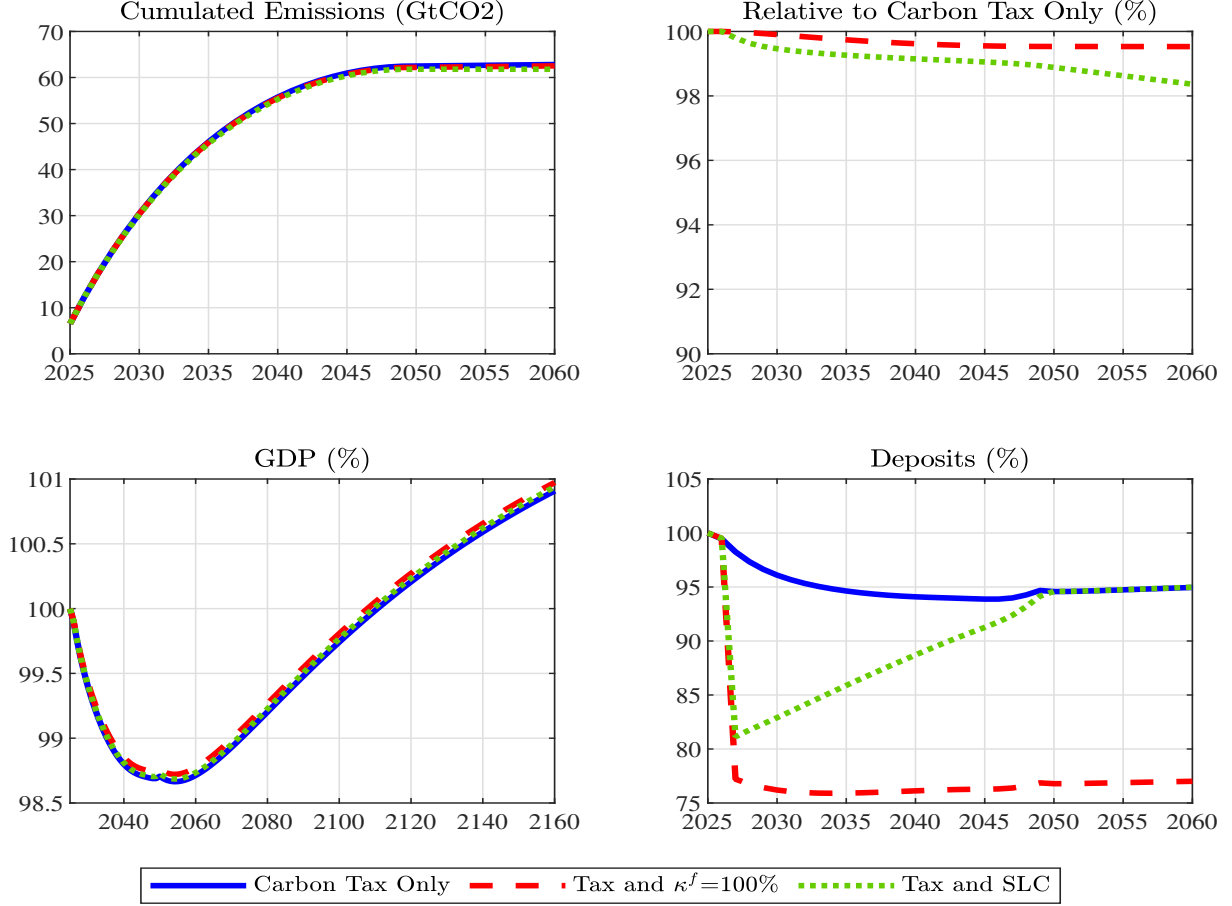


Figure 4: Transition Path: Cumulated carbon dioxide emissions (GtCO₂) since 2025 for different policies. The solid black line reflects a perfect foresight transition to net zero induced linear emission tax path, starting from 0\$/tCO₂ in 2025 to 150\$/tCO₂ in 2050. Adding fossil-penalizing capital requirements and sustainability-linked capital requirements to the policy mix is reflected by the dashed red and dotted green line, respectively. The dashed blue line refers to the case without any climate policy.

requirement always has a (small) negative effect on the return on fossil capital throughout the transition. Nevertheless, our model implies that the effect on world emissions would be rather modest as well: world emissions amounted to 33 GtCO₂ in 2022 alone. We have shown that differentiated bank capital requirements have a macroeconomically relevant but quantitatively small effect on aggregate emissions and this effect is more pronounced if they reward emission reduction activities.

4.4 Robustness

As a last step, we show that our results are robust to changing parameters in the climate block and the sectoral shares in the production sector. This is motivated by the considerable uncertainty surrounding in particular the parameters governing climate damages and the cost of abating emissions. By contrast, the sectoral shares are quite heterogeneous across countries. We focus on a comparison between fossil capital requirements, sustainability-linked capital requirements, a modest emission tax and the emission tax τ^e consistent with net zero. For each policy, we report the fossil energy share, the emission reduction relative to the baseline without

policies, the liquidity premium as an easily interpretable measure of deposit scarcity and, most importantly, the welfare gain relative to the baseline.

High Abatement Cost					
	Baseline	$\kappa^f = 100\%$	SLC	$\tau^e = 3\$/\text{tCO}_2$	Net zero τ^e
Fossil Energy Share	80.00%	79.17%	76.69%	73.71%	0%
Δ Emissions	-	-0.96%	-3.94%	-8.04%	-100%
Bank Failure Prob	0.54%	<0.1%	<0.1%	0.54%	0.54%
Liquidity Premium	-41bps	-60bps	-59bps	-42bps	-45bps
Welfare Gain	-	-0.10%	+0.36%	+0.95%	+4.16%
High Climate Damages					
Fossil Energy Share	80.00%	79.18%	76.13%	72.36%	0%
Δ Emissions	-	-0.93%	-4.52%	-9.46%	-100%
Bank Failure Prob	0.54%	<0.1%	<0.1%	0.54%	0.54%
Liquidity Premium	-41bps	-60bps	-59bps	-41bps	-44bps
Welfare Gain	-	-0.04%	+0.79%	+1.88%	+15.07%
High Energy Share					
Fossil Energy Share	80.00%	79.03%	75.71%	72.36%	0%
Δ Emissions	-	-1.00%	-4.94%	-9.69%	-100%
Bank Failure Prob	0.54%	<0.1%	<0.1%	0.54%	0.54%
Liquidity Premium	-39bps	-70bps	-67bps	-39bps	-42bps
Welfare Gain	-	-0.22%	+0.67%	+1.68%	+8.52%
High Initial Fossil Share					
Fossil Energy Share	85.00%	84.39%	81.07%	77.07%	0%
Δ Emissions	-	-0.74%	-4.47%	-9.50%	-100%
Bank Failure Prob	0.54%	<0.1%	<0.1%	0.54%	0.54%
Liquidity Premium	-41bps	-60bps	-61bps	-42bps	-44bps
Welfare Gain	-	-0.14%	+0.46%	+1.22%	+7.31%

Table 4: Robustness: Long Run Effects of Selected Policies. We compare the baseline model without emission taxes and symmetric capital requirements (column 1) to different climate policies. Column 2 refers to a simple fossil-penalizing capital requirement ($\kappa^f = 1$), column 3 to a sustainability-linked capital requirement with $\kappa^{pen} = 1$ and $\kappa^{reward} = 0.08$, column 4 to a $3\$/\text{tCO}_2$ emission tax, and column 5 to the emission tax consistent with net zero. The emission change and welfare are computed relative to the baseline. The fossil energy share is defined relative to total energy production $(1 - a_t^*)z_t^f/z_t^e$. The bank failure rate and the liquidity premium $r_t^D - r_t$ are annualized. Welfare is defined in Equation (1), computed relative to the baseline without climate policies and expressed in consumption equivalents.

In the first panel of Table 4, we report the case with higher abatement costs $\theta_1 = 0.07$, such that an emission tax of $149\$/\text{tCO}_2$ is required to achieve net zero. Consequently, the SLC and the $3\$/\text{tCO}_2$ tax have a smaller effect on the fossil energy share and the emission reduction. Notably, the effects on the liquidity premium are quite similar to the baseline, such that the net welfare effect of any climate policy is smaller than in the baseline calibration.

As a second robustness, we increase the climate damage parameter Ψ_E to from 0.1 to 0.15, which implies that damages associated with unmitigated global warming amount to 5% of GDP, compared to 3% in the baseline calibration. The efficacy of different climate policies is very

similar to the baseline, the small differences stem from the larger long run damages from emissions that depress aggregate TFP. As the effects on liquidity provision are similar to the baseline, the welfare effects of all policies are much larger.

The other two robustness concern the sectoral structure in the production side. First, we consider a higher energy share in the production function. We increase $\tilde{\nu}$ such that it implies an energy share of 15%, compared to 10% used in the baseline calibration. Within the energy sector, we leave the share of fossil energy firms constant. All climate policies have stronger effects, both on emission reduction and liquidity provision, as the energy sector has a larger macroeconomic impact. Second, we impose a larger share of firms initially operating the fossil technology, i.e. we increase ν to imply an initial fossil share of 85%, compared to 80% in the baseline calibration. This slightly exacerbates the adverse effects of all climate policies on liquidity provision. In both cases, however, the welfare effects are slightly more pronounced, but of similar magnitude to the baseline.

5 Optimal Bank Regulation

In this section, we study optimal bank capital regulation in the presence of emission taxes. In the main text, we focus on the long run, while Section C.4 displays optimal bank regulation along the net zero transition induced by emission taxes. The macroeconomic effects of a permanent emission tax on key welfare relevant variables are summarized in Figure 5. The top row shows that the abatement share increases in the emission tax. The effect is larger if the tax level is low, i.e. fossil energy firms pick the "low-hanging fruits" when it comes to de-carbonizing the energy sector. This follows from the convex specification of abatement costs $\frac{\theta_1}{\theta_2+1} a_t^{\theta_2+1} z_t^f$.

At the same time, emission taxes have a negative effect on aggregate investment. Importantly, the reduction of bank balance sheets in response to a permanently lower aggregate loan demand implies a smaller supply of deposits to households and the deposit spread widens by around 2bps. Compared to the large decline in available deposits induced by fossil capital requirements, the adverse side effects of emission taxes on the financial sector are comparatively small. Furthermore, in Figure C.2, we show that the *long run* bank failure rate does not depend on the emission tax as the bank capital requirement is always binding. Summing up, the net welfare effect is nevertheless positive and increasing in τ_t^e and quantitatively large.

How do the macroeconomic effect of emission taxes affect optimal bank capital regulation? Figure 6 shows how a change in the symmetric capital requirement κ^{sym} affects macroeconomic aggregates without emission taxes (dashed red line) and with net zero emission taxes (dotted green line). The upper row shows that, irrespective of the climate policy regime in place, capital requirements directly affect bank failure and corporate default rates and liquidity provision to households. Importantly, the transmission of bank regulation does not depend on the climate policy regime in place.¹¹ Specifically, the top right panel demonstrates that tighter requirements reduce the failure probability of banks. Furthermore, by reducing banks' loan supply, tight

¹¹For example, an increase in κ^{sym} from 8% to 9% reduces the bank failure rate by around 0.3 percentage points and the liquidity premium by one basis point.

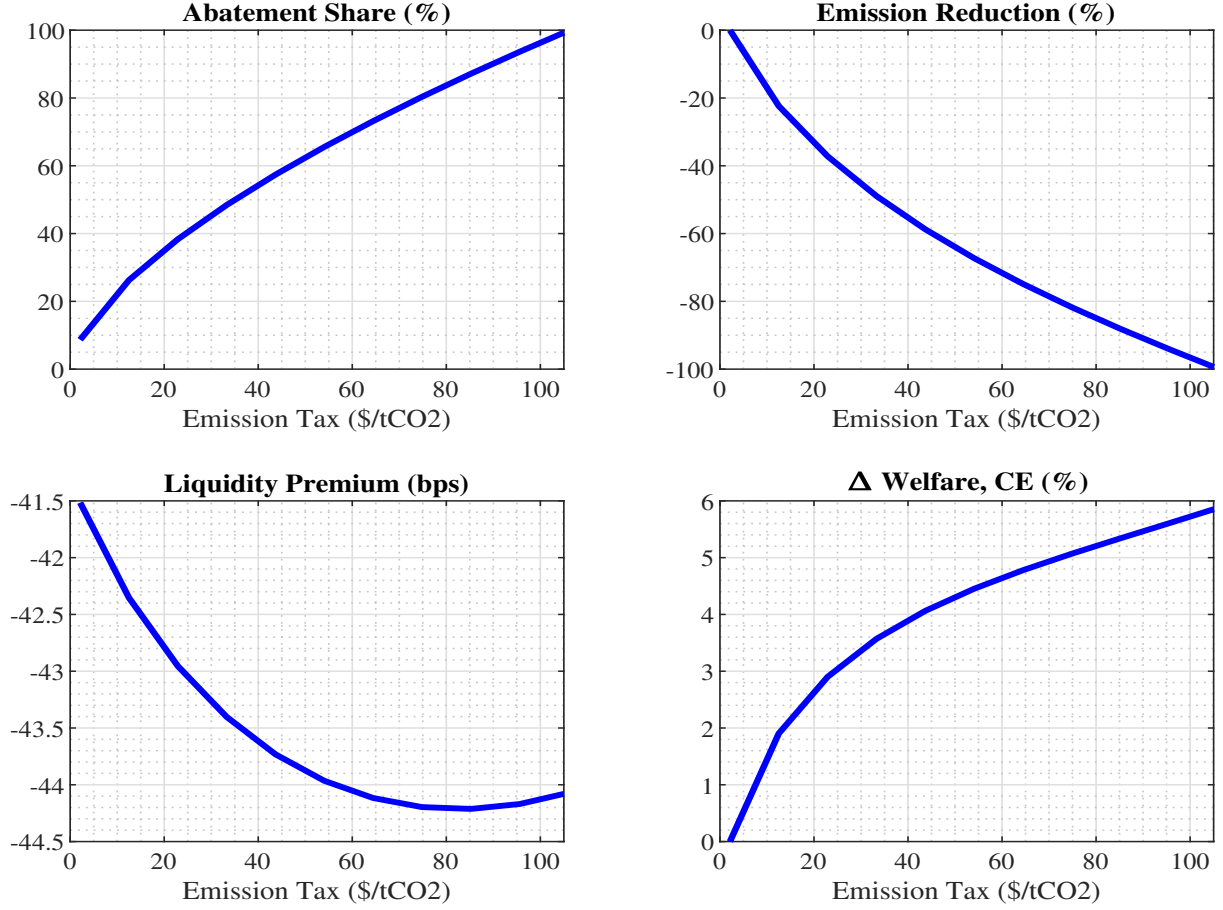


Figure 5: Long Run Effects of Emission Taxes. This figure displays the effects of long run emission taxes on key welfare relevant aggregates: Share of abatement emissions a_t^* (in %, top left), the emission reduction relative to the benchmark without emission taxes (in %, top right), and the liquidity premium $r_t^D - r_t^f$ (annualized in basis points, bottom left). The welfare change is computed relative to the baseline model with symmetric capital requirements of 8% and without emission taxes (bottom right). The emission tax on the x-axis is expressed in \$/tCO2.

capital requirements also reduce risk-taking and default risk in the non-financial sector. Again, the deadweight losses corporate default do not substantially depend on the climate policy regime in place. At the same time, the liquidity premium increases in the capital requirements since deposits become scarcer and, thus, more valuable to households (top left panel). This represents the key trade-off for bank capital regulation in the long run.

We have seen that higher emission taxes reduce liquidity provision through a contraction in aggregate loan demand, but leave the long run bank failure rate unchanged. Thus, the trade-off is clearly resolved in favor of liquidity provision: The bottom row shows that it is optimal to slightly decrease the capital requirement from 8% to 7.4%. This counteracts the adverse effect of emission taxes on liquidity provision.¹² The effect of modifying either the climate block or the sectoral structure on optimal capital regulation is negligible, as the welfare-maximizing capital requirement under net zero would still be 7.4% in all robustness checks outlined in Section 4.4.

¹²In Figure C.3, we show how other welfare-relevant aggregates, such as consumption and the corporate default rate are affected by bank capital regulation.

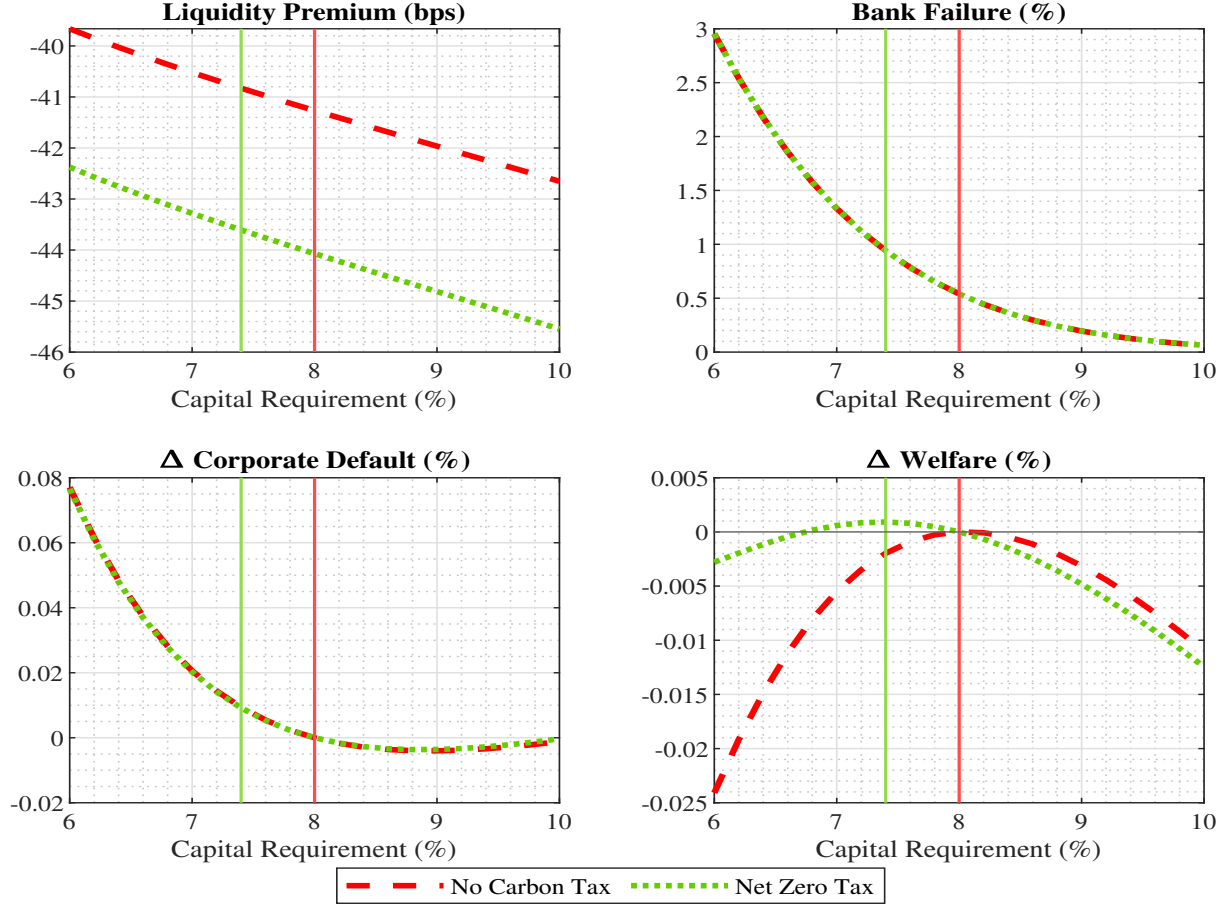


Figure 6: Long Run Effects of Symmetric Capital Requirements: This figure displays the effects of long run capital requirements on key welfare relevant aggregates, such as the liquidity premium $r_t^D - r_t^f$ (annualized in basis points, top left), the bank failure rate $F(\bar{\mu})$ (in %, top right), the corporate default rate, normalized to the benchmark with an 8% capital requirement (in %, bottom left). Welfare (bottom right) is expressed in certainty equivalents relative to the baseline model without emission taxes ($\tau_t^e = 0$) and an 8% capital requirement. The capital requirement on the x-axis is expressed in percent. The vertical lines represent the optimal capital requirement without emission taxes (dashed red line) and with the net zero emission tax (dotted green line).

6 Conclusion

In this paper, we have proposed a multi-sector DSGE model with two layers of default to study optimal bank capital regulation and climate policy in a joint framework. We show that differentiated capital requirements for clean and fossil loans have a macroeconomically relevant but quantitatively small effect on emissions. The effects are larger if fossil capital requirements reward abatement effort in the fossil energy sector but still fall way short of contributing to a meaningful transition to net zero. Furthermore, such policies have negative effects on deposit supply to households, which can render them detrimental to welfare. These results are robust to reasonable variations in the parameters governing climate policy and the sectoral structure of the economy. Our model also provides a useful laboratory to study implications of the net zero transition for optimal bank capital regulation. We show that optimal bank capital requirements are relaxed symmetrically across sectors along the net zero transition to counter negative deposit supply effects.

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Online Appendix for "Climate Change and the Macroeconomics of Bank Capital Regulation"

February 15, 2025

A Model Appendix

This section provides additional analytical steps to derive banks' loan supply as well as firms' loan demand and investment.

Banks: Funding Choice The profit maximization problem of the representative bank is given by

$$\max_{d_{t+1}, l_{t+1}^s} d_{t+1} - \sum_s q(\bar{m}_{t+1}^s) l_{t+1}^s + \mathbb{E}_t \left[\Lambda_{t,t+1} \int_{\bar{\mu}_{t+1}}^{\infty} \mu_{t+1} \sum_s \mathcal{R}_{t+1}^s l_{t+1}^s - (1 + r_t^D) \frac{d_{t+1}}{\pi_{t+1}} dF(\mu_{t+1}) \right] .$$

where profits are discounted using the household sdf $\Lambda_{t,t+1}$. Raising deposits increases bank dividends in period t by one unit. This exceeds expected discounted repayment obligations in period $t + 1$:

$$\mathbb{E}_t \left[\Lambda_{t,t+1} \frac{1 + r_t^D}{\pi_{t+1}} (1 - F(\bar{\mu}_{t+1})) \right] < 1 ,$$

where $F(\bar{\mu}_{t+1}) \equiv \int_0^{\bar{\mu}_{t+1}} dF(\mu_{t+1})$ denotes the bank failure probability. First, this follows from the liquidity services that deposits provide to households, where we obtain $\Lambda_{t,t+1} \frac{1+r_t^D}{\pi_{t+1}} < 1$ from combining the household first-order conditions for deposits (2) and the definition of the risk-free rate (3). Furthermore, the risk of bank failure combined with the bailout guarantee implies that raising one unit of deposits in period t only involves an expected real repayment of $1 - F(\bar{\mu}_{t+1}) < 1$ units in $t + 1$, whenever $\mu_{t+1} > 0$. This implies that the capital requirement binds in all states. Note that bank capital requirements have an effect on loan supply if only one of the two frictions is present.

Banks: Credit Supply. Next, we plug the (binding) bank equity constraint (6) into the maximization problem:

$$\begin{aligned} \max_{l_{t+1}^s} & \frac{\pi_{t+1}}{1 + r_t^D} \sum_s (1 - \kappa_t^s) \mathbb{E}_t[\mathcal{R}_{t+1}^s] l_{t+1}^s - \sum_s q(\bar{m}_{t+1}^s) l_{t+1}^s \\ & + \mathbb{E}_t \left[\Lambda_{t,t+1} \int_{\bar{\mu}_{t+1}}^{\infty} \mu_{t+1} \sum_s \mathcal{R}_{t+1}^s l_{t+1}^s - (1 - \kappa_t^s) \mathcal{R}_{t+1}^s l_{t+1}^s dF(\mu_{t+1}) \right] , \end{aligned}$$

and differentiate w.r.t. l_{t+1}^s to get

$$0 = \frac{\pi_{t+1}(1 - \kappa_t^s)\mathbb{E}_t[\mathcal{R}_{t+1}^s]}{1 + r_t^D} - q(\bar{m}_{t+1}^s) + \mathbb{E}_t \left[\Lambda_{t,t+1} \left\{ (1 - G(\bar{\mu}_{t+1}))\mathcal{R}_{t+1}^s - (1 - F(\bar{\mu}_{t+1}))(1 - \kappa_t^s)\mathcal{R}_{t+1}^s \right\} \right] = 0.$$

Rearranging for $q(\bar{m}_{t+1}^s)$ yields the loan price schedule (7):

$$q(\bar{m}_{t+1}^s) = \mathbb{E}_t \left[\left((1 - \kappa_t^s)\Xi_{t+1} + \bar{\Lambda}_{t,t+1} \right) \mathcal{R}_{t+1}^s \right],$$

where

$$\bar{\Lambda}_{t,t+1} \equiv \Lambda_{t,t+1} \left(1 - G(\bar{\mu}_{t+1}) \right) \quad \text{and} \quad \Xi_{t+1} \equiv \frac{\pi_{t+1}}{1 + r_t^D} - \Lambda_{t,t+1} (1 - F(\bar{\mu}_{t+1}))$$

The derivative of the loan price schedule with respect to the risk choice $\bar{m}_{t+1}^s$ follows:

$$q'(\bar{m}_{t+1}^s) = \mathbb{E}_t \left[\left((1 - \kappa_t^s)\Xi_{t+1} + \bar{\Lambda}_{t,t+1} \right) \left(-\frac{\chi G(\bar{m}_{t+1}^s)}{\pi_{t+1}(\bar{m}_{t+1}^s)^2} - \frac{\chi \varphi F'(\bar{m}_{t+1}^s)}{\pi_{t+1}} + (1 - \chi) \frac{\partial \bar{m}_{t+2}^s}{\partial \bar{m}_{t+1}^s} q'(\bar{m}_{t+2}^s) \right) \right]. \quad (\text{A.1})$$

Policy Function for Leverage. Characterizing the loan and investment choices in the intermediate good sector needs to take the stickiness of firm leverage into account, which arises in the model since debt is long-term. This implies that next period's risk choice $\bar{m}_{t+2}^s$ depends on the current risk choice $\bar{m}_{t+1}^s$. This dependency affects both the risk choice (18) and the derivative of the loan price (A.1).

Following Gomes et al. (2016), we pin down $\frac{\partial \bar{m}_{t+2}^s}{\partial \bar{m}_{t+1}^s}$, which is an object that depends on the unknown policy function for risk choice, by using an additional condition. Note that this can be obtained by further differentiating the first order condition for loans with respect to $\bar{m}_{t+1}^s$. From (16), we get an expression for the Lagrangian multiplier:

$$\lambda_t^s = \frac{l_{t+1}^s}{\bar{m}_{t+1}^s} q(\bar{m}_{t+1}^s) - \tilde{\Lambda}_{t,t+1} \frac{l_{t+1}^s}{\bar{m}_{t+1}^s} \left[\chi(1 - F(\bar{m}_{t+1}^s)) + (1 - \chi)q(\bar{m}_{t+2}^s) \right].$$

By differentiating this w.r.t. $\bar{m}_{t+1}$, we obtain

$$q(\bar{m}_{t+1}^s) + \bar{m}_{t+1}^s q'(\bar{m}_{t+1}^s) = \frac{\tilde{\Lambda}_{t,t+1}}{\pi_{t+1}} \left[\chi(1 - F(\bar{m}_{t+1}^s)) + (1 - \chi) \left(q(\bar{m}_{t+2}^s) + q'(\bar{m}_{t+2}^s) \bar{m}_{t+1}^s \frac{\partial \bar{m}_{t+2}^s}{\partial \bar{m}_{t+1}^s} \right) \right], \quad (\text{A.2})$$

Final Good Firms. The cost minimization problem of final good firms yields the following standard first-order conditions:

$$mc_t \alpha \tilde{\nu} \frac{y_t}{\tilde{z}_t} \left(\frac{\tilde{z}_t}{z_t^e} \right)^{\frac{1}{\epsilon}} \left(\frac{z_t^e}{z_t^c} \right)^{\frac{1}{\epsilon}} = p_t^c, \quad (\text{A.3})$$

$$mc_t \alpha \tilde{\nu} (1 - \nu) \frac{y_t}{\tilde{z}_t} \left(\frac{\tilde{z}_t}{z_t^e} \right)^{\frac{1}{\epsilon}} \left(\frac{z_t^e}{z_t^f} \right)^{\frac{1}{\epsilon}} = p_t^f, \quad (\text{A.4})$$

$$mc_t \alpha (1 - \tilde{\nu}) \frac{y_t}{\tilde{z}_t} \left(\frac{\tilde{z}_t}{z_t^n} \right)^{\frac{1}{\epsilon}} = p_t^n, \quad (\text{A.5})$$

$$mc_t (1 - \alpha) \frac{y_t}{n_t} = w_t, \quad (\text{A.6})$$

where mc_t is the real marginal cost of production for the final good. Denoting with ϕ the elasticity of substitution across final goods, final good monopolists face price rigidities à la Rotemberg, with Ψ_P being the parameter governing the degree of nominal rigidity. The price-setting maximization problem of final good producer i is then given by

$$\max \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{c_t^{-\gamma_C}}{c_0^{-\gamma_C}} \left\{ \left(\frac{P_t(i)}{P_t} \right)^{-\phi} \left(\frac{P_t(i)}{P_t} - mc_t \right) y_t - \frac{\Psi_P}{2} \left(\frac{P_t(i)}{P_{t-1}(i)} \right)^{-\phi} \left(\frac{P_t(i)}{P_t} - 1 \right)^2 y_t \right\} \right].$$

Solving the maximization problem and imposing symmetry, we arrive at the standard New Keynesian Philips curve

$$\mathbb{E}_t \left[\Lambda_{t,t+1} \frac{y_{t+1}}{y_t} (\pi_{t+1} - 1) \pi_{t+1} \right] + \frac{\phi}{\Psi_P} \left(mc_t - \frac{\phi - 1}{\phi} \right) = (\pi_t - 1) \pi_t. \quad (\text{A.7})$$

Investment Good Firms. There is a representative producer for each of the three investment goods that intermediate firms acquire at price ψ_t^s . To produce one unit of each investment good, these firms use $(1 + \frac{\Psi_I}{2} (\frac{i_t^s}{i_{t-1}^s}))$ units of the final good. The profit maximization problem

$$\max_{\{i_t^s\}_{t=0}^{\infty}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{c_t^{-\gamma_C}}{c_0^{-\gamma_C}} \left\{ \psi_t^s i_t^s - \left(1 + \frac{\Psi_I}{2} \left(\frac{i_t^s}{i_{t-1}^s} - 1 \right)^2 \right) i_t^s \right\} \right]$$

yields the first-order condition for (type-specific) investment good supply

$$\psi_t^s = 1 + \frac{\Psi_I}{2} \left(\frac{i_t^s}{i_{t-1}^s} - 1 \right)^2 + \Psi_I \left(\frac{i_t^s}{i_{t-1}^s} - 1 \right) \frac{i_t^s}{i_{t-1}^s} - \mathbb{E}_t \left[\Lambda_{t,t+1} \Psi_I \left(\frac{i_{t+1}^s}{i_t^s} - 1 \right) \left(\frac{i_{t+1}^s}{i_t^s} \right)^2 \right]. \quad (\text{A.8})$$

A.1 Simplified Model

In the simplified case with one period debt, the multiplier on the risk choice reduces to $\lambda_{t+1}^f = -l_{t+1}^f q'(\bar{m}_{t+1}^f)$ and the default threshold becomes $\bar{m}_{t+1}^f = \frac{l_{t+1}^f}{k_{t+1}^f (p_{t+1}^f - \xi_{t+1})}$. Exploiting this together with full depreciation in the first order condition for investment (17) yields

$$1 + q'(\bar{m}_{t+1}^f) (\bar{m}_{t+1}^f)^2 (p_{t+1}^f - \xi_{t+1}) = \tilde{\beta} \left((p_{t+1}^f - \xi_{t+1}) (1 - G(\bar{m}_{t+1}^f)) \right),$$

which is equation (26). The fossil energy price on the right hand side is multiplied by the expected profitability conditional on not defaulting $1 - G(\bar{m}_{t+1}^f)$. This makes investment into fossil capital less attractive. The left-hand side contains an expression related to the pricing of loans: more investment makes default less likely in the next period, which ceteris paribus increases the loan price and makes fossil capital more attractive. Quantitatively, it turns out that the expected firm productivity conditional on defaulting $G(\bar{m}_{t+1}^f)$ is close to zero.

Without restructuring costs but a full loss of output in default, we obtain an analytically tractable expression for the loan price schedule and its derivative. Specifically, we have

$$q(\bar{m}_{t+1}^f) = \left((1 - \kappa_t^f) \Xi_{t+1} + \bar{\Lambda}_{t,t+1} \right) \left(1 - F(\bar{m}_{t+1}^f) \right), \quad (\text{A.9})$$

where $\bar{\Lambda}_{t,t+1} = \beta(1 - G(\mu_{t+1}))$ is the bank-owner sdf. Plugging (A.9) and $G(\bar{m}_{t+1}^f) \approx 0$ into the first-order condition for investment (26) and rearranging, we obtain

$$1 = \tilde{\beta} \left\{ \left((1 - \kappa_t^f) \Xi_{t+1} + \bar{\Lambda}_{t,t+1} \right) f(\bar{m}_{t+1}^f) (\bar{m}_{t+1}^f)^2 + 1 \right\} (p_{t+1}^f - \xi_{t+1}). \quad (\text{A.10})$$

We can simplify this further by using that also both the expected bank portfolio return conditional on failure $G(\bar{\mu}_{t+1})$ is close to zero. Then, the bank-owner sdf approximately satisfies $\bar{\Lambda}_{t,t+1} = \beta(1 - G(\bar{\mu}_{t+1})) \approx \beta$.

Abstracting from non-energy intermediate goods and labor, there is an equally simple fossil energy demand function $p_t^f = (1 - \nu) \left(\frac{y_t}{z_t^f} \right)^{1/\epsilon}$, which obtains from setting $\alpha = \tilde{\nu} = 1$ in equation (A.4) in Section A. We impose market clearing and plug the demand for fossil energy into the first-order condition for fossil capital (A.10) to arrive at (27).

B Data Sources

Table B.1 summarizes our data sources. Whenever data is available, we focus on the period 1984-2013 and obtain data at annual frequency. Leverage is available at quarterly frequency until 2012 from the replication data provided by Gomes et al. (2016). To obtain an annual time series, we simply average corporate leverage ratios within each year. Corporate default rates are also provided by Gomes et al. (2016) and we use the default frequency of all rated firms to construct our data moments. Recovery rates are provided by the World Bank, following the methodology outlined in Djankov et al. (2008). We take the simple time series average for the US, which is however only available after 2013. Data for a set of developed OECD countries (Australia, Austria, Canada, Germany, Denmark, Finland, France, Netherlands, Norway, South Korea, Sweden, United Kingdom) is available since 2003 and average across countries and years also yields a value of 82%.

Table B.1: Data: Sources and Ticker

Variable	Source	Series/Ticker
Bank Failure Rate	FDIC	Total Commercial Banks Failures: Assisted Merger Failures: Paid Off
Corporate Default	Gomes et al (2016)	All Rated Firms
Corporate Leverage	Gomes et al (2016)	Net Debt over Net Assets
Corporate Recovery Rate	World Bank	Recovery rate (cents on the dollar)
Real GDP	St Louis Fed	GDPCA
Nominal GDP	St Louis Fed	NGDPXDCUSA
Nominal Deposits	St Louis Fed	SAVINGSSL
Inflation	St Louis Fed	FPCPITOTLZGUSA
Federal Funds Rate	St Louis Fed	RIFSPFFNA

In Figure B.1, we plot key financial market variables against the cyclical component of real GDP. Corporate leverage ratios, default rates, and bank failure rates show a strong comovement with each other and tend to peak around recessions. This is consistent with the counter-cyclical behavior of these variables in our model, see Table 2. Furthermore, the time-series volatilities of key financial market variables are of a comparable magnitude to the volatility of cyclical GDP.

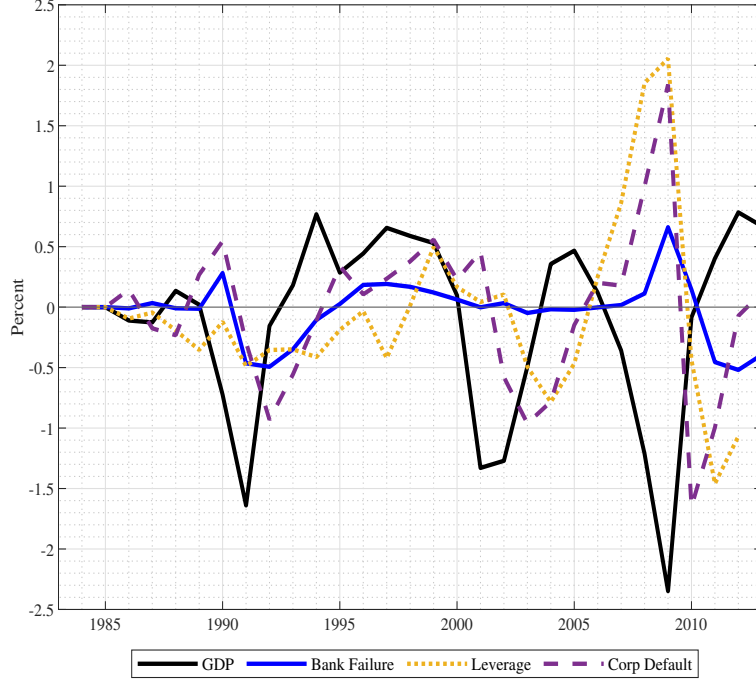


Figure B.1: Bank Failure and Corporate Default Rates: Annual real GDP is logged and de-trended using a HP-filter with smoothing parameter 6.25.

C Additional Numerical Results

In this section, we provide additional numerical results from subjecting our baseline model to different capital requirements and emission taxes.

C.1 Fossil and Sustainability-Linked Capital Requirements

Figure C.1 displays additional important sectoral and aggregate effects of sector-specific bank capital regulation. This complements the discussion of key welfare relevant aggregates in Figure 1. Specifically, we vary the simple fossil capital requirement (dashed red line) and the penalty capital requirement κ_t^{pen} , while keeping $\kappa_t^{reward} = \kappa_t^c = 0.08$ (dotted green line).

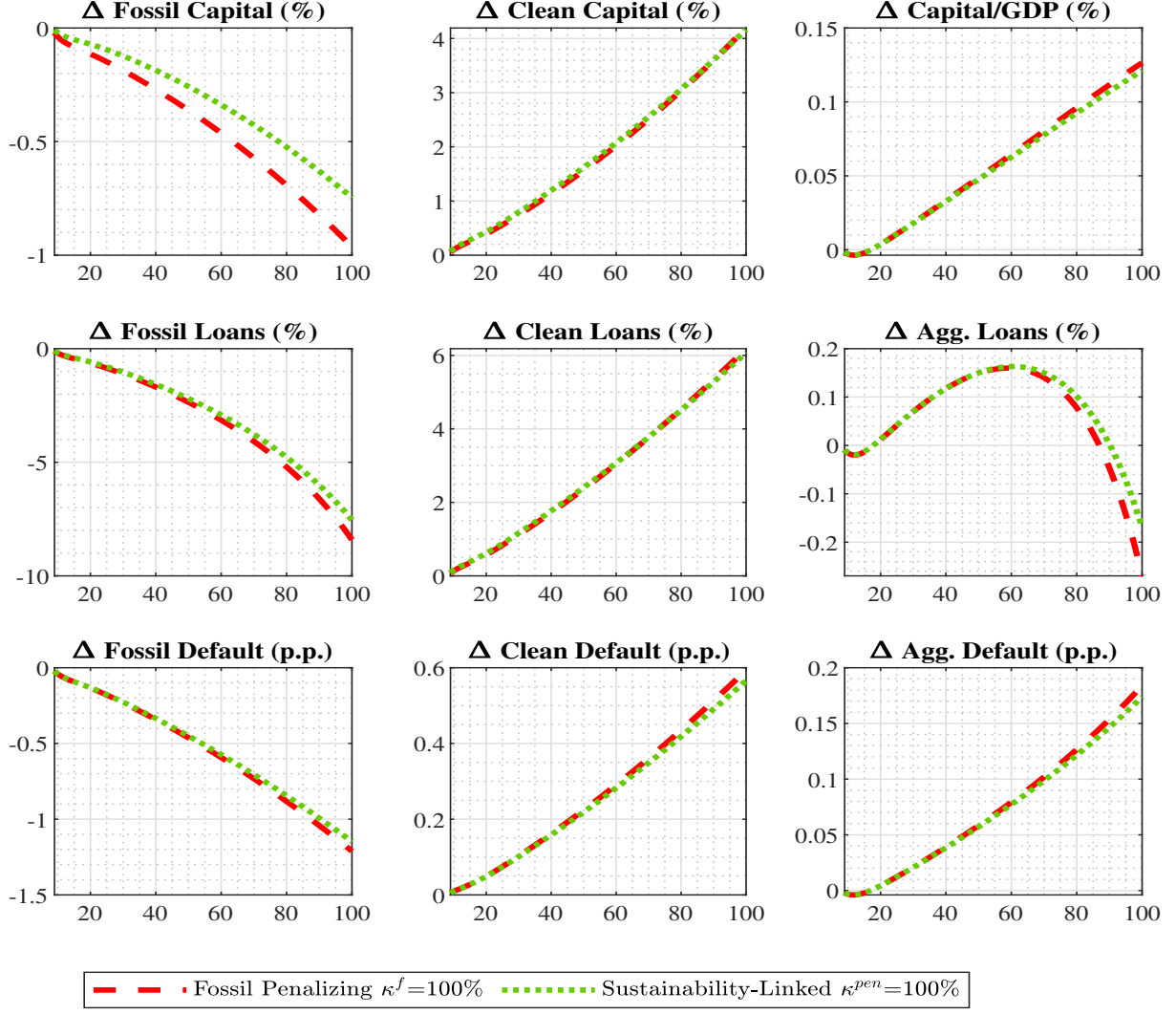


Figure C.1: Sectoral Effects of Fossil Capital Requirements. This figure displays additional welfare relevant aggregates in the top row. The annualized bank failure rate is expressed in percent, consumption and GDP are expressed relative to the baseline without emission taxes. The second, third and fourth row compare the sectoral and aggregate effects of emission taxes in terms of investment, loans and default rates (annualized in percent). The emission tax on the x-axis is expressed in \$/tCO₂.

C.2 Emission Taxes

In this section, we show how additional endogenous variables are affected by emission taxes in the long run. Furthermore, we decompose the macroeconomic effects of emission taxes (Figure 5) into their sectoral origins. The top row of Figure C.2 shows that consumption and GDP increase if climate policy is more stringent. This is due to reduction in damages associated with the emission externality. At the same time, the bank failure rate is independent of the long run climate policy regime in place. The reason is that, in our model, the regulator effectively chooses the failure rate by setting capital requirements, which we hold constant in this experiment. This is not to say that the *shift* towards stringent emission taxes comes without disruptions to the financial sector. We discuss these in Section C.4.

Note that there is an equilibrium effect operating against the reduction in bank deposits

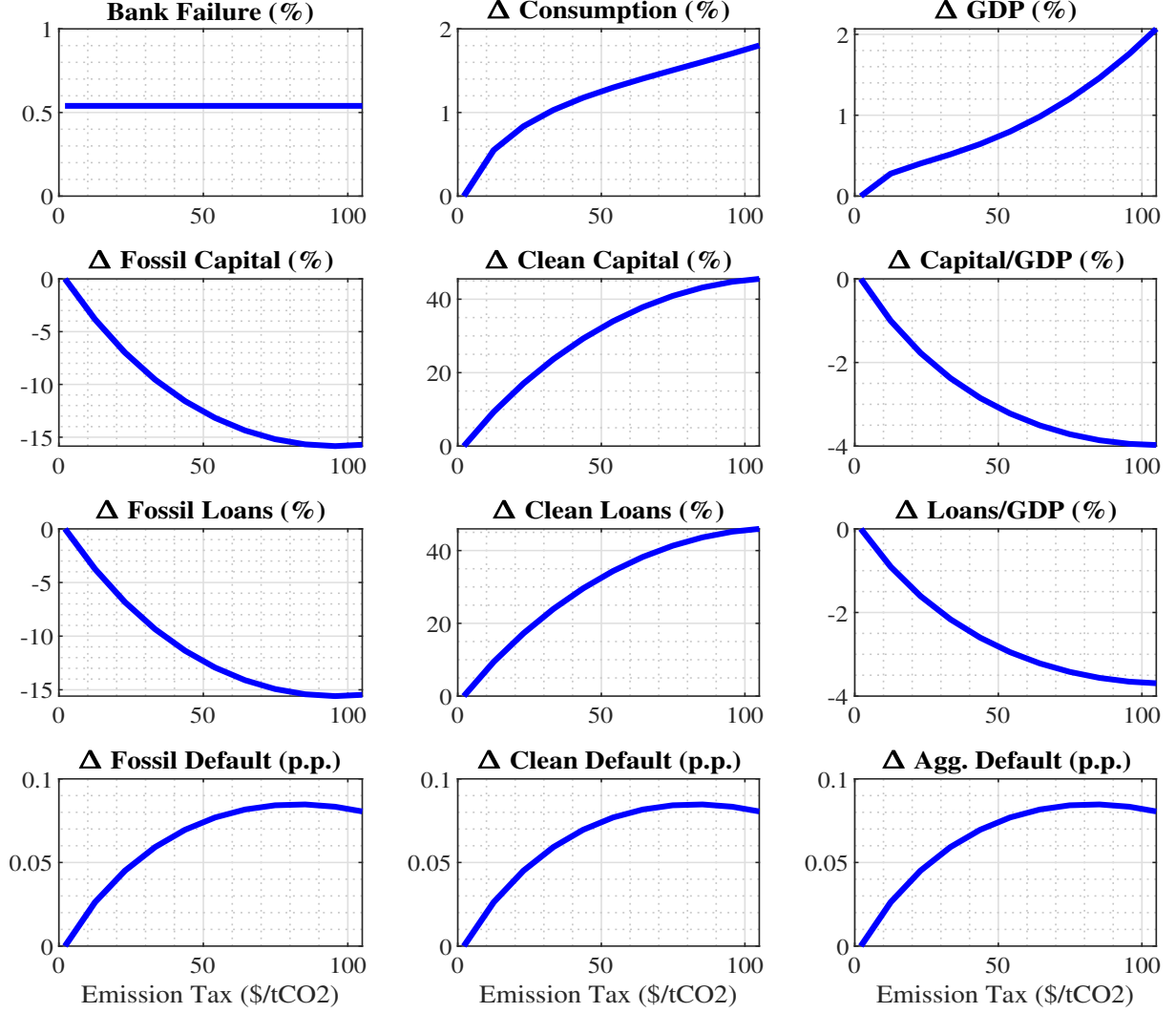


Figure C.2: Additional Long Run Effects of Emission Taxes. This figure displays additional welfare relevant aggregates in the top row. The annualized bank failure rate is expressed in percent, consumption and GDP are expressed relative to the baseline without emission taxes. The second, third and fourth row compare the sectoral and aggregate effects of emission taxes in terms of investment, loans and default rates (annualized in percent). The emission tax on the x-axis is expressed in \$/tCO2.

induced by a smaller loan demand. By increasing deposit scarcity, the liquidity premium over the risk free rate widens, which depresses bank funding costs and increases loan supply. This increases corporate default rates for all firms by almost 0.1 percentage points, which is non-negligible from a macroeconomic perspective given the baseline level of 2.7%.

C.3 Symmetric Capital Requirements

In the main text, we have shown how symmetric bank capital requirements affect the key welfare-relevant macroeconomic aggregates: the bank failure rate, the liquidity premium measuring the scarcity of deposits, and socially harmful emissions. Figure C.3 shows that lenient capital requirements induce an increase in GDP, aggregate capital and loans which follow from the expansionary effect of lenient capital requirements on banks' credit supply. At the same time,

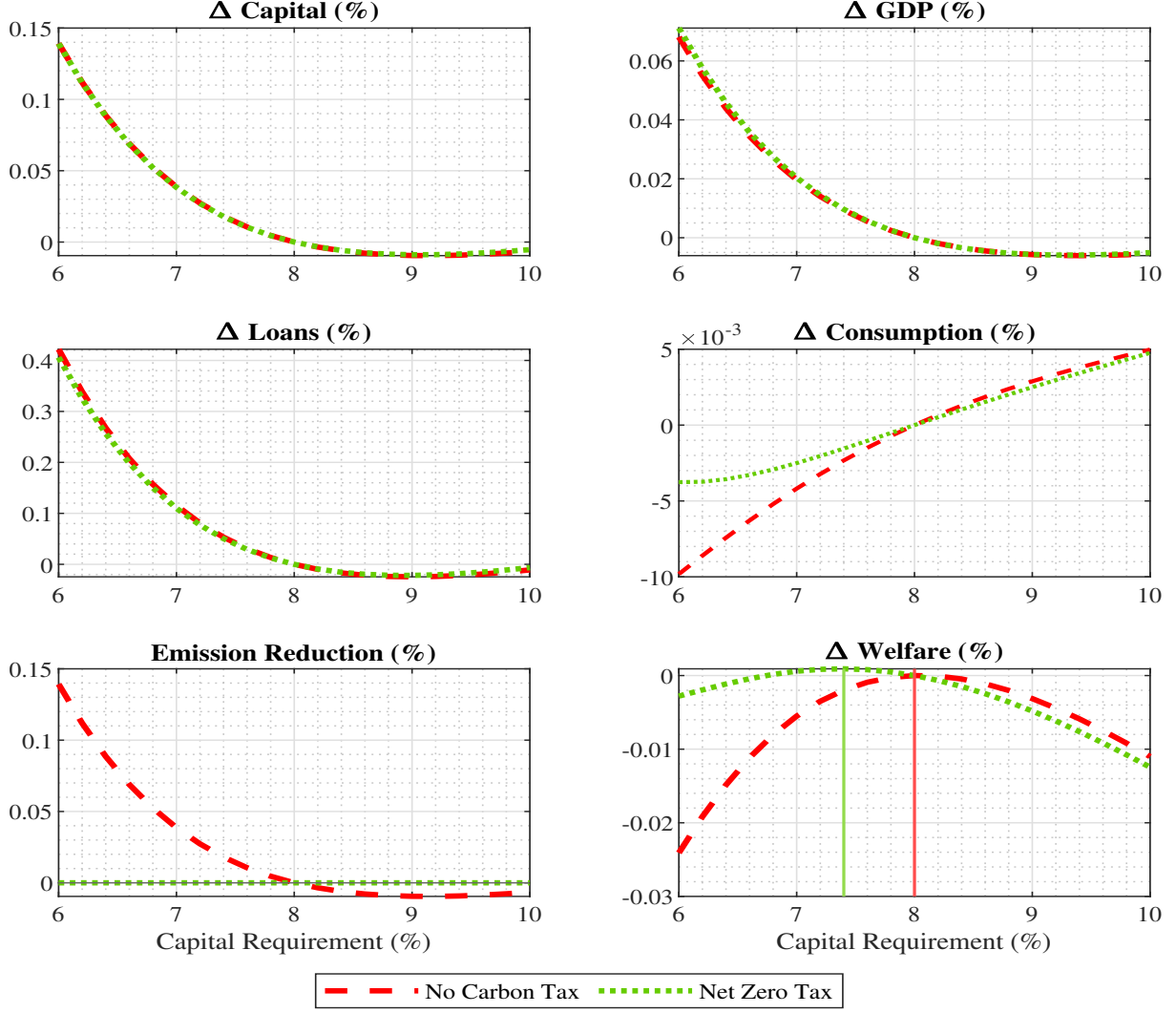


Figure C.3: Additional Long Run Effects of Symmetric Capital Requirements. This figure displays additional welfare relevant aggregates in the top row. The annualized bank failure rate is expressed in percent, consumption and GDP are expressed relative to the baseline without emission taxes. The second, third and fourth row compare the sectoral and aggregate effects of emission taxes in terms of investment, loans and default rates (annualized in percent). The symmetric capital requirement on the x-axis is expressed in %.

aggregate consumption declines despite the expansionary effect on economy activity. This reflects, on the one hand the larger share of resources devoted to restructuring banks, which is a direct implication of a higher bank failure rate.

In addition to the resources losses of higher bank failure rate, risk-taking in the non-financial sector is endogenous in our model. In the bottom left panel of Figure C.3, we show how changes to the capital requirement affect firm risk-taking. If capital requirements are high κ_t^{sym} , loan supply contracts and firm's optimal capital structure is tilted towards equity. The corporate default rate falls. This is a consequence of smaller weight $(1 - \kappa_t^{sym})$ on the deposit-financing wedge Ξ_{t+1} in banks' loan pricing condition (16). By contrast, more lenient capital requirements increase default risk in the banking and non-financial sector.

The bottom right panel shows that there is an interior solution to the policy trade-off between

liquidity provision to households and resource losses from corporate default and bank failure. How does the climate policy regime affect this trade-off? As we have seen before, emission taxes have no effect on bank failure rates in the long run and only rather modest effects on corporate default (see the lower panel of Figure C.2), but they have adverse side effect on liquidity provision to households, through their negative effects on loan demand and the size of bank balance sheets. Thus, the optimal bank capital requirement is slightly smaller.

C.4 Optimal Bank Regulation and the Net Zero Transition

In this section, we show the implications of a net zero transition across sectors and in the aggregate, complementing our analysis of transition dynamics in Figure 4. The economy is solved under perfect foresight for an emission tax that linearly increases from zero to its full abatement level in 2050. We first consider capital requirements fixed at their initial level of $\kappa^{sym} = 8\%$. The results are displayed by solid black lines in Figure C.4. We also study bank capital requirements that are linearly adjusted along the net zero transition such that they reach their optimal long run level of 7.4% in 2050. This is shown by the dashed red lines.

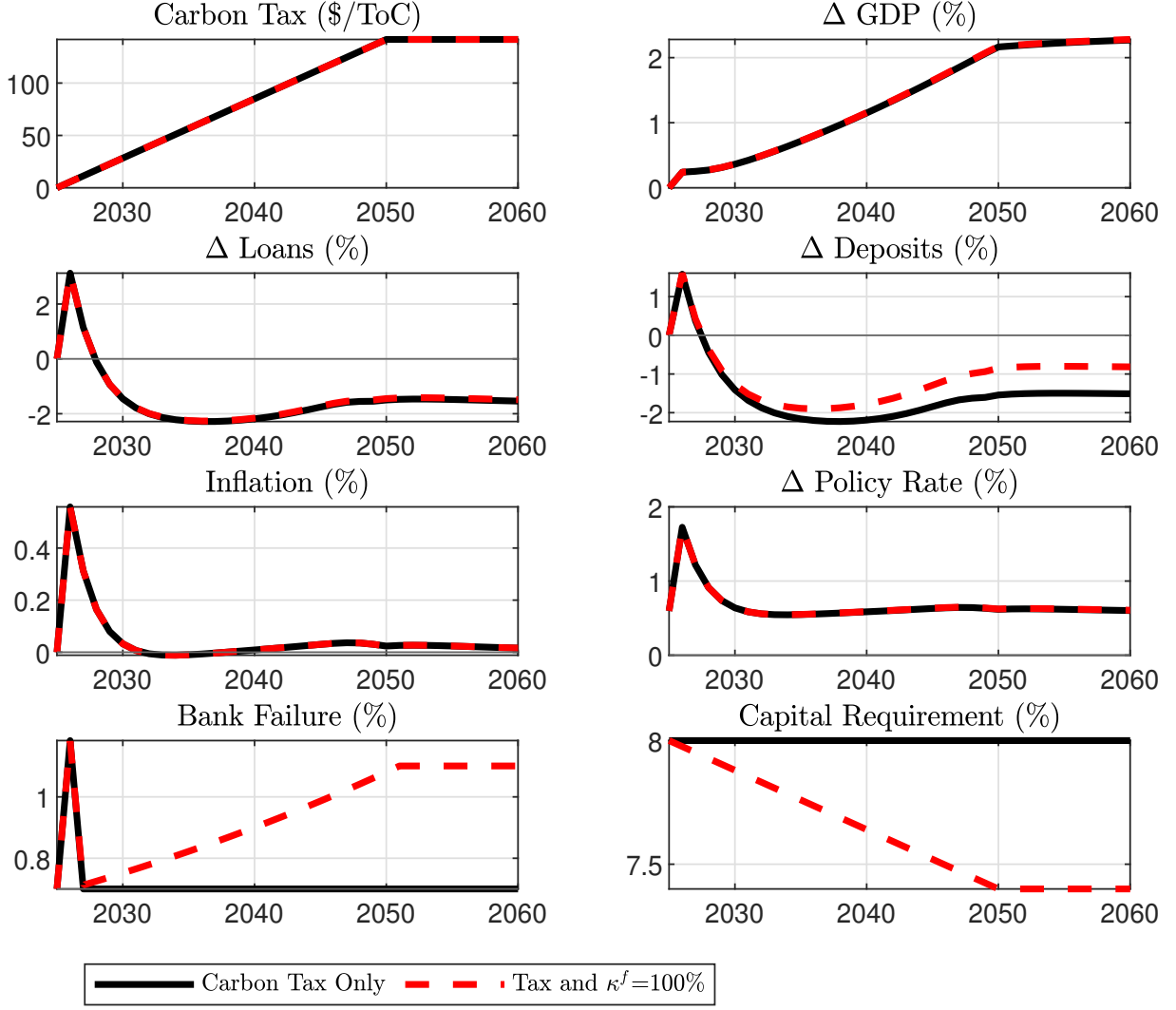
The first row of Figure C.4 shows how emission taxes translate into emission abatement in the fossil energy sector. Costly abatement is the only way for fossil energy firms to reduce emissions without downsizing their assets. The relative compliance cost ξ_t/p_t^f , which measure the policy-induced wedge for fossil energy firms, increase from zero to almost twenty percent at the full abatement tax. As the second row of Figure C.4 shows, the emission tax path induces a substantial shift from fossil towards clean investment. Specifically, the clean sector increases its investment by around 50%, while investment declines by more than 20% in the fossil sector. Thus, emission taxes directly affect the sectoral composition of the intermediate good sector.

The response of sector-specific default risk is more nuanced. We can distinguish between *impact* and *short run* effects. In the announcement period, the default threshold of fossil energy firms ($\bar{m}_t^f \equiv \frac{\chi l_t^f}{(p_t^f - \xi_t)k_t^f}$) increases due to the unexpected spike in compliance cost ξ_t . Their default rate increases on impact from 1.8% to 2.4%. By contrast, there is a slight drop in the default rate of clean energy firms: intermediate good firms substitute away from fossil energy, which increases the price of clean energy and, thereby, also the clean default threshold.

Once firms can adjust their loan issuance and investment in the short run, their leverage (third row) and their default rate (fourth row) are tied to their risk choice. Here, we observe that the increase in clean investment is accompanied by risk-taking in the clean sector. Conversely, the fossil sector exhibits a drop in its leverage while its default rate is slightly below the clean default rate. Once the transition is complete, both firm types choose the same leverage ratio, since the risk choice is solely determined by the cost and benefits of debt issuance.

Emission taxes are directly relevant to fossil energy firms which experience an increase in their default rate on impact. Due to an equilibrium effect operating through the demand for different intermediate goods, clean energy firms experience an increase in their revenues such that their default rate declines on impact. The aggregate effect then depends on the strength of this equilibrium effect. It turns out that aggregate default risk goes up, as sector-specific

Figure C.4: Net Zero Transition: Optimal Bank Regulation



Notes: Perfect foresight transition: linear emission tax path from zero in 2025 to 150\$/ToC in 2050. All endogenous variables are computed while keeping capital requirements at their symmetric baseline value.

default risk is a convex function $F(\bar{m}_t^s)$ of the default threshold. The log-normal distributional assumption on idiosyncratic revenues ensures that $f'(\bar{m}_t^s)$ is positive, i.e. the increase in fossil default risk exceeds the decline in clean default risk. As a consequence of these sectoral effects, the bank failure rate increases because the realized return on their loan portfolio experiences a negative shock, see the lower panel of Figure C.4. Due to the assumption of binding capital requirements, banks raise equity from households and their default rate reverts back to steady state immediately.

How is the loan market affected along the transition? Intermediate goods are imperfectly substitutable in the final good production technology (19). The negative shock to the productivity of fossil energy firms is, thus, recessionary: GDP and loan demand fall. By the bank balance sheet, aggregate deposits decline as well, such that deposits become scarcer. This is summarized

by the first two rows of Figure C.4. The scarcity of deposits makes them more valuable, such that banks can refinance themselves at favorable conditions. From the loan pricing condition (7), we observe that banks pass on their financing conditions to firms: credit supply increases and the relative attractiveness of debt financing increases for all sectors. By the first-order condition for leverage (18), firms take more risk, which is reflected in a larger corporate default rate in the aggregate along the entire transition path and in the long run equilibrium. Figure C.4 provides additional details on these sector-specific transition dynamics.